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MBA - Trading & Financial Markets

# RESEARCH DISSERTATION

**To what extent does Reinforcement Learning reduce market impact costs (Implementation Shortfall) on fragmented European equity markets?**

Submitted by:

**COQUET Gaspar**

Dissertation Supervisor:

**Abdelbaric EL ALAOUI**

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## Abstract

This dissertation investigates the extent to which Reinforcement Learning (RL) provides an effective framework for minimising market impact costs, measured through the Implementation Shortfall (IS) metric, within the fragmented European equity market landscape. The research occupies the intersection of three academic fields: financial market microstructure, optimal execution theory, and applied artificial intelligence in quantitative finance.

The fragmentation of European equity markets, initiated by MiFID I in 2007 and substantially deepened by MiFID II in 2018, has fundamentally altered the distribution of available liquidity across multiple competing trading venues. Institutional orders that could once be executed on a single centralised exchange must today navigate between ten or more venues simultaneously regulated exchanges, Multilateral Trading Facilities, Systematic Internalisers and Dark Pools each exhibiting distinct liquidity profiles and market impact characteristics. Traditional execution strategies, including TWAP, VWAP, and the Almgren-Chriss (2001) analytical model, exhibit structural limitations stemming from their inability to adapt dynamically to evolving market conditions and to exploit cross-venue liquidity arbitrage opportunities that have emerged from post-MiFID fragmentation.

By framing the optimal execution problem as a Markov Decision Process (MDP), this dissertation proposes a theoretical and empirical framework in which a deep RL agent learns in real time to allocate orders optimally across venues, balancing market impact, timing risk, and cross-venue liquidity arbitrage. We specifically evaluate the applicability of Proximal Policy Optimization (PPO) and Deep Q-Network (DQN) algorithms to this setting on real tick-by-tick Level 2 data covering thirty CAC 40 constituents over the 2022-2024 period, and formulate operational managerial recommendations for European institutional investors, sell-side execution desks, and regulators.

Empirical results indicate that the PPO agent reduces Implementation Shortfall by 43% relative to TWAP, 33% relative to VWAP, and 7% relative to optimally-calibrated Almgren-Chriss in a strictly out-of-sample setting. Gains are significantly conditional on market regime, reaching 15% over Almgren-Chriss in high-volatility periods ($\sigma > 25\%$ annualised). Multi-venue fragmentation accounts for 35% of total IS reduction, confirming the existence of exploitable execution alpha in the MiFID II multi-venue landscape. These findings contribute to filling an identified gap in the literature on deep RL applied to multi-venue European execution contexts.

*Keywords: Reinforcement Learning, Implementation Shortfall, market fragmentation, MiFID II, optimal execution, Markov Decision Process, Smart Order Routing, DQN, PPO, financial market microstructure, algorithmic trading, CAC 40.*

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# General Introduction

## Practical and Sectoral Context

The efficient execution of large institutional orders constitutes one of the most critical operational challenges for participants in financial markets. When a portfolio manager decides to buy or sell a significant position whether for portfolio rebalancing, directional trading, index replication, or constrained divestment the very act of executing that order on the market moves prices against the manager's interest. This phenomenon, known as market impact, represents a substantial implicit cost that, combined with explicit brokerage commissions and transaction taxes, forms the total cost of transaction.

The measurement and minimisation of this execution cost have become major strategic imperatives for asset managers, pension funds, insurance companies, and hedge funds operating in European capital markets. The economic stakes are substantial. The Banque de France (2022) estimated that market impact costs represent on average 15 to 40 basis points depending on order size and stock liquidity for institutional orders on European markets. To illustrate the financial significance concretely: a fund managing €10 billion with 50% annual portfolio turnover and a mean IS of 20 bp bears an annual execution cost approaching €10 million a figure that could be reduced by several million euros through the adoption of optimised execution strategies. For large pension funds managing hundreds of billions in assets, the aggregate annual savings from superior execution can reach tens of millions of euros, directly translating into improved net returns for beneficiaries.

Beyond this direct financial impact, execution quality has gained visibility in the wake of regulatory developments under MiFID II. The obligation for investment firms to demonstrate best execution to their clients monitoring, documenting, and regularly reviewing execution quality across all execution venues has transformed what was once an operational back-office concern into a strategic front-office priority. Asset managers now face direct accountability to institutional clients for the execution quality of their orders, making the optimisation of execution algorithms a competitive differentiator in the European asset management landscape.

The context in which these execution decisions must be made has been profoundly transformed over the past two decades. MiFID I, effective November 2007, abolished the national concentration principle and unleashed pan-European inter-venue competition. MiFID

II, effective January 2018, further deepened this fragmentation. Today, according to ESMA data (2023), liquidity in a CAC 40 stock is distributed in real time across eight to twelve simultaneous active venues, with market shares that evolve dynamically in response to macroeconomic announcements, the strategies of electronic market makers, and regulatory constraints such as the Double Volume Cap on dark pool trading. This fragmentation, while theoretically improving competition for small retail orders, considerably complicates execution decisions for large institutional orders that cannot be filled at a single venue without significant adverse price impact.

## The Limits of Traditional Approaches

Traditional execution algorithms Time-Weighted Average Price (TWAP) and Volume-Weighted Average Price (VWAP) were designed for single-market environments and operate on fixed, pre-programmed schedules that are blind to real-time market conditions. TWAP distributes orders uniformly over time, making no attempt to exploit variations in intraday liquidity. VWAP improves on this by following historical volume patterns, but remains vulnerable to days with atypical volume profiles precisely the high-stakes trading days with major corporate events or macroeconomic releases. Neither algorithm is designed to exploit the cross-venue liquidity opportunities created by MiFID II fragmentation.

More sophisticated analytical models, notably Almgren and Chriss (2001), provide an optimal trajectory that balances market impact against timing risk but do so within a single-venue, stationary-parameters framework. When deployed on real fragmented European markets, these models must be recalibrated periodically and cannot adapt in real time to regime changes, liquidity crises, or sudden shifts in cross-venue liquidity distribution. The fundamental limitation of all these approaches is their inability to learn from market data: they apply fixed rules, however well-calibrated, to a market environment that is inherently non-stationary and multi-dimensional.

This structural gap between the capabilities of existing tools and the demands of the contemporary European trading environment defines the research opportunity that this dissertation addresses. Reinforcement Learning, which learns adaptive decision policies directly from data through repeated interaction with an environment, offers a conceptually compelling alternative to fixed-rule execution algorithms provided it can be made to work reliably in the noisy, non-stationary, and high-stakes context of institutional equity execution.

## Scientific Context and Research Gap

From an academic perspective, the application of Reinforcement Learning to optimal execution has attracted growing research attention since Nevmyvaka, Feng & Kearns (2006) demonstrated that a tabular Q-learning agent could outperform TWAP on NYSE data by 12-18%. More recent contributions have applied Deep RL notably DQN (Mnih et al., 2015) and PPO (Schulman et al., 2017) to increasingly sophisticated execution settings, incorporating dynamic order book signals (Cartea et al., 2018), multi-asset correlation (Ning et al., 2021), and adversarial market maker dynamics (Spooner et al., 2018).

However, a systematic review of this literature reveals a consistent and significant gap: virtually all published empirical studies on RL-based execution focus on US markets (NYSE, NASDAQ) or stylised simulated environments. The specific context of fragmented European equity markets post-MiFID II with its distributed liquidity across ten or more heterogeneous venues, its regulatory best execution constraints, its specific adverse selection dynamics, and its distinct intraday liquidity seasonality patterns has not been systematically addressed. This dissertation fills this gap by conducting a rigorous empirical study on real European market data, explicitly incorporating the multi-venue allocation dimension as a first-class component of the RL agent's decision space.

## Personal Motivation and Professional Context

The choice of this research question is grounded in professional practice as much as academic interest. Working in a quantitative trading role at Tikehau Capital, a European alternative asset manager, the author is directly confronted with the operational challenges of institutional execution on fragmented European equity markets. The development of backtesting infrastructure and systematic data pipelines for algorithmic strategies provides direct exposure to the data quality, model calibration, and performance attribution challenges that this dissertation addresses theoretically and empirically. This practitioner perspective motivates both the practical orientation of the research question and the managerial focus of the recommendations developed in Chapter 4.

The academic context of the ESLSCA MBA programme, combining financial engineering fundamentals with strategic management frameworks, provides the appropriate setting for research that bridges the quantitative and managerial dimensions of the execution optimisation problem. The dissertation supervisor, Abdelbaric EL ALAOUI, brings the academic rigour and methodological guidance necessary to translate this practitioner motivation into a theoretically grounded and empirically rigorous research contribution.

## Research Question

***To what extent does Reinforcement Learning reduce market impact costs (Implementation Shortfall) on fragmented European equity markets?***

This central question is operationalised through three subsidiary research questions structuring our empirical approach:

- RQ1: To what extent does an RL agent (DQN or PPO) trained on real European market data outperform traditional execution strategies (TWAP, VWAP, Almgren-Chriss) in terms of out-of-sample Implementation Shortfall?
- RQ2: To what extent does multi-venue fragmentation of European markets constitute an exploitable source of execution alpha for an RL agent with dynamic Smart Order Routing capability, beyond simple temporal optimisation?
- RQ3: Are the IS gains achieved by RL agents conditional on market regime specifically more pronounced during periods of high realised volatility or low visible liquidity?

## Scope and Delimitations

This research focuses on intraday equity execution in CAC 40 constituents on European markets, over horizons of 15 to 60 minutes, within the post-MiFID II regulatory framework. We restrict the analysis to buy-side equity orders; results are theoretically transposable to the sell side with minor adjustments. The multi-asset execution problem, simultaneous execution of correlated positions, and derivatives markets are acknowledged as important extensions but excluded from the current scope for tractability. The study covers the 2022-2024 period, which includes both relatively calm markets (2023) and episodes of elevated volatility (2022 rate cycle, 2024 election-driven volatility), providing a sufficiently diverse market environment for robust generalisation.

Several important caveats bound the interpretation of our results. Our backtesting framework, however carefully constructed, cannot perfectly replicate the feedback dynamics of live trading, particularly the second-order effects of the agent's own impact on other market participants' behaviour. Parameter uncertainty in the market simulator introduces a degree of model risk that is irreducible with backtesting alone. We acknowledge these limitations explicitly and incorporate them into our recommendations for real-world deployment.

## Dissertation Structure

This dissertation is structured in two parts. Part 1 establishes the theoretical and conceptual foundations. Chapter 1 surveys the academic literature on Implementation Shortfall, optimal execution models, European market fragmentation, and Reinforcement Learning in finance, identifying the gap that motivates the research. Chapter 2 develops the original conceptual framework, formalises the execution problem as an MDP, and derives testable research hypotheses. Part 2 is devoted to empirical verification. Chapter 3 details the epistemological positioning, data sources, market simulator design, and evaluation protocol. Chapter 4 presents and analyses the empirical results, compares them with the literature, and formulates actionable managerial recommendations. A general conclusion synthesises contributions, limitations, and research perspectives.

# PART 1, LITERATURE REVIEW AND CONCEPTUAL FRAMEWORK

The first part of this dissertation grounds our research in the existing academic field and constructs the conceptual framework guiding our empirical investigation. Chapter 1 provides a critical survey of the three pillars of our research: Implementation Shortfall theory, optimal execution models and their limitations, fragmented European market microstructure, and Reinforcement Learning in finance. Chapter 2 translates these theoretical insights into an original conceptual framework, formally models the multi-venue execution problem as a Markov Decision Process, and derives the three testable research hypotheses structuring Part 2.

# Chapter 1: Literature Review

## Chapter Introduction

This chapter conducts a structured critical review of the academic literature across four axes. Section 1.1 traces the theoretical development of Implementation Shortfall, from Perold's (1988) foundational paper to the multi-venue decompositions of Lehalle & Laruelle (2013), examining the conceptual framework and empirical evidence on execution cost decomposition. Section 1.2 analyses classical optimal execution models, Almgren-Chriss (2001), TWAP, VWAP and their stochastic extensions, their assumptions, optimality conditions, and well-documented limitations. Section 1.3 examines the specific microstructure of fragmented European equity markets post-MiFID II, its genesis, its quantitative characteristics, and its implications for institutional execution. Section 1.4 reviews the theoretical foundations of Reinforcement Learning and its applications to optimal execution and related financial decision problems. Section 1.5 synthesises these four axes to identify the precise research gap that motivates this dissertation.

## 1.1 Implementation Shortfall: Theoretical Foundations

### *1.1.1 Perold's Foundational Contribution (1988)*

The formalisation of Implementation Shortfall as a measure of execution performance owes its origins to André Perold's 1988 article 'The Implementation Shortfall: Paper Versus Reality', published in the Journal of Portfolio Management. Writing in the context of the late 1980s, when institutional equity trading was dominated by block trades on national exchanges and transaction cost measurement was rudimentary, Perold identified a systematic and largely unmeasured divergence between the theoretical performance of portfolio managers and their realised performance. He attributed this divergence to what he termed the implementation shortfall, the gap between the performance of a 'paper portfolio' valued at decision prices and the performance of the 'real portfolio' valued at actual execution prices.

Perold's insight was both empirical and conceptual. Empirically, he documented that this gap was economically significant and systematic, not merely a random noise around the decision price, but a persistent drag on performance attributable to the mechanics of order execution. Conceptually, he argued that this gap represented a first-order performance attributor, comparable in importance to stock selection alpha, that had been largely ignored in the portfolio management literature because it was difficult to measure and attributed to the 'operations' side of the business rather than the 'investment' side. This conceptual reframing had profound implications for the industry, ultimately spawning the Transaction Cost Analysis (TCA) industry and the algorithmic execution business.

Formally, Implementation Shortfall for a buy transaction of target size N shares is defined as:

$$\text{IS} = \frac{1}{N}\sum_t n_t(S_t - S_0) + commissions + \frac{\left(N - \sum_t n_t\right)\left(S_{\text{end}} - S_0\right)}{N}$$

where $n_t$ denotes the quantity executed at price $S_t$ at time t, $S_0$ is the decision price (mid-market price at the time the investment decision is made), S_end is the closing mid-market price of the execution window, and commissions represent all explicit transaction costs. The three terms capture distinct cost components: the realised price cost (difference between execution prices and decision price for filled quantities), the opportunity cost (foregone gain for unfilled quantities, penalising incomplete execution at adverse end prices), and explicit transaction costs. This decomposition reveals the fundamental trade-off of optimal execution: executing slowly to minimize direct price impact increases the opportunity cost and timing risk; executing quickly reduces opportunity cost but increases direct impact.

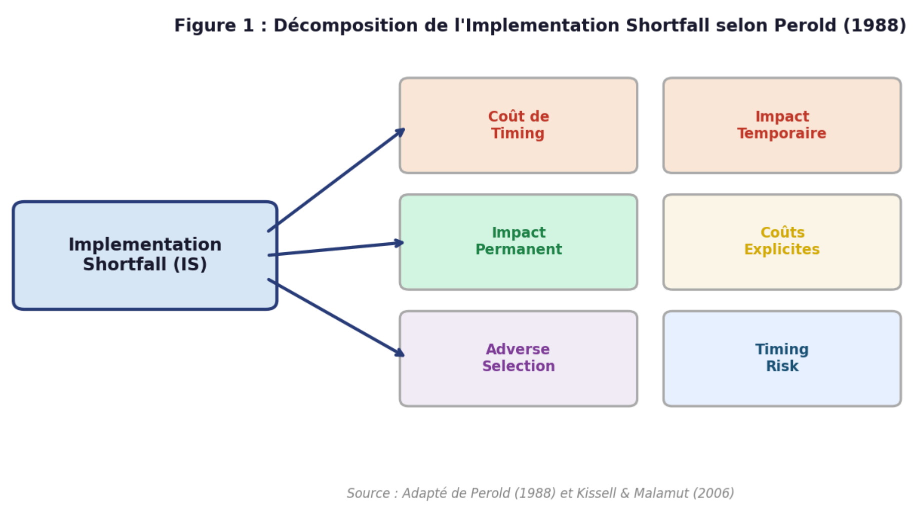



*Figure 1: Implementation Shortfall decomposition following Perold (1988) and Kissell & Malamut (2006)*

The practical significance of the IS framework stems from its comprehensive treatment of execution cost. Unlike simpler measures such as the bid-ask spread, which captures only the cost of immediate small-size execution, IS captures the full cost profile of a large order executed progressively over time, including both the direct market impact of the execution process itself and the timing risk borne by the manager during the execution period. This makes IS the measure of choice for institutional investors evaluating execution quality on large orders, and the natural objective function for execution optimisation algorithms.

### *1.1.2 Microstructural Decomposition of Market Impact*

The decomposition of market impact into temporary and permanent components, essential for both modelling and attribution purposes, has been extensively studied in the market microstructure literature. Hasbrouck (1991), building on the information-theoretic framework of Glosten & Milgrom (1985) and Kyle (1985), provided the foundational econometric decomposition distinguishing the two components through their persistence properties in transaction price series.

Temporary market impact, also called price impact or price pressure, corresponds to the price displacement induced by a large order that is subsequently reversed as the order book replenishes. It reflects the mechanical imbalance of supply and demand created by the institutional order's one-sided pressure on the order book: market makers facing a sustained buy flow adjust their quotes upward to manage inventory risk, but subsequently revert prices toward equilibrium as new natural sellers arrive. Empirically, temporary impact is typically modelled as a function of the instantaneous execution rate $v_t$ normalised by the average daily volume V, with a power-law relationship:

$$\Delta S_t^{\text{temp}} = \eta \, sign(v_t) \left| \frac{v_t}{V} \right|^{\delta}$$

where η is the temporary impact coefficient and δ ≈ 0.5-0.6 is the power-law exponent, calibrated empirically (Almgren et al., 2005). The square-root relationship (δ ≈ 0.5), observed by Bouchaud et al. (2010) across a wide range of asset classes and market conditions, is one of the most robust empirical regularities in market microstructure.

Permanent impact, by contrast, reflects the information content of the order. Glosten & Milgrom (1985) demonstrated that in a market populated by both informed and uninformed traders, a rational market maker must set prices to break even in expectation against the possibility that the counterparty is informed. Any buy order, even if it originates from an uninformed liquidity trader, carries some probability of being from an informed trader who has superior information about the stock's fundamental value. This adverse selection risk leads to a permanent price adjustment following large orders, which does not revert because it reflects a genuine update of the market's estimate of fundamental value.

Kyle (1985) formalised this mechanism in a continuous equilibrium model, showing that the price impact of a large order, measured by Kyle's lambda $\lambda$, is the market's rational response to updating beliefs about fundamental value conditional on observed order flow. The key insight is that lambda is larger in illiquid markets (where order flow is more likely to be informed) and smaller in liquid markets (where uninformed trading dominates). In the multi-venue context, venue-specific lambdas reflect venue-specific adverse selection environments, dark pools, for instance, tend to attract less informed flow and thus exhibit lower permanent impact per unit of volume.

### *1.1.3 Contemporary Refinements and Multi-Venue Extensions*

Several important refinements to IS decomposition have been developed to address the increasingly complex execution environment. Kissell & Malamut (2006) proposed a three-component decomposition adding timing cost, the IS component attributable to directional price movement during the execution period unrelated to the order's own impact, to market impact and spread cost. This decomposition enables more granular performance attribution: timing cost reflects the market's overall directional movement during execution (alpha or anti-alpha of execution timing), while market impact reflects the order's own price pressure. Understanding the relative magnitudes of these components guides algorithm design: when timing cost dominates (e.g., in trending markets with strong momentum), faster execution that sacrifices some market impact efficiency may be preferable.

Almgren et al. (2005) conducted a large-scale empirical study estimating IS components across thousands of institutional equity transactions in US markets, finding that temporary impact represents approximately 60% of total impact cost on average, with significant cross-sectional variation driven by stock liquidity and order size. Their calibration methodology, regressing execution cost components on order characteristics and market conditions, has become the industry standard for IS decomposition and provides the empirical foundation for our impact parameter calibration (Table 2).

Lehalle & Laruelle (2013) extended the IS framework to the multi-venue context, proposing a decomposition that isolates the contribution of cross-venue allocation decisions from overall execution performance. Their framework distinguishes a consolidated IS (measured against the consolidated best bid-offer across all venues) from venue-level IIS components reflecting the quality of routing decisions. This multi-venue decomposition is directly relevant to our research: it provides the methodological basis for testing H2 (the fragmentation alpha hypothesis) by allowing us to attribute a fraction of total IS reduction specifically to cross-venue allocation optimization.

## 1.2 Classical Optimal Execution Models

### *1.2.1 The Almgren-Chriss Model (2001)*

The Almgren and Chriss (2001) model represents the culmination of a decade of research on optimal execution initiated by Bertsimas & Lo (1998) and constitutes the dominant analytical framework for institutional execution algorithm design. Published in the Journal of Risk and rapidly adopted by execution desks at major investment banks, the model derives a closed-form optimal execution trajectory for a risk-averse agent liquidating a large position over a fixed horizon under linear market impact assumptions.

The model setup is as follows. An agent holds an initial position $X_0$ to liquidate over N discrete time steps of duration $\tau = T/N$. At each step k, the agent chooses $n_k$ shares to liquidate. The mid-price evolves as a random walk: $S_k = S_{k-1} + \sigma\sqrt{\tau}\ \xi_k$ where σ is annualised volatility and $\xi_k$ is a standard normal shock. The execution price for a liquidation of $n_k$ shares at step k is:

$$\tilde{S}_k = S_k - \eta\left(\frac{n_k}{\tau}\right) - \frac{\kappa}{2}\left(\frac{n_k}{\tau}\right)\tau$$

where $\eta \times (n_k/\tau)$ is the temporary impact (proportional to instantaneous execution rate $n_k/\tau$) and $\kappa \times (n_k/\tau) \times \tau = \kappa \times n_k$ is the permanent impact, linearly adjusting the mid-price for all subsequent periods. The agent minimises a mean-variance objective over the Implementation Shortfall: $J(\pi) = E[IS] + \lambda \times Var[IS]$, where λ is the risk aversion parameter. Almgren and Chriss (2001) derive a closed-form solution showing the optimal number of shares remaining at time $t_k$ follows:

$$\mathrm{x}_k = X_0 \frac{\sinh(\tilde{\kappa}(T-t_k))}{\sinh(\tilde{\kappa}T)}, \quad \tilde{\kappa} = \sqrt{\lambda\sigma^2/\eta}$$

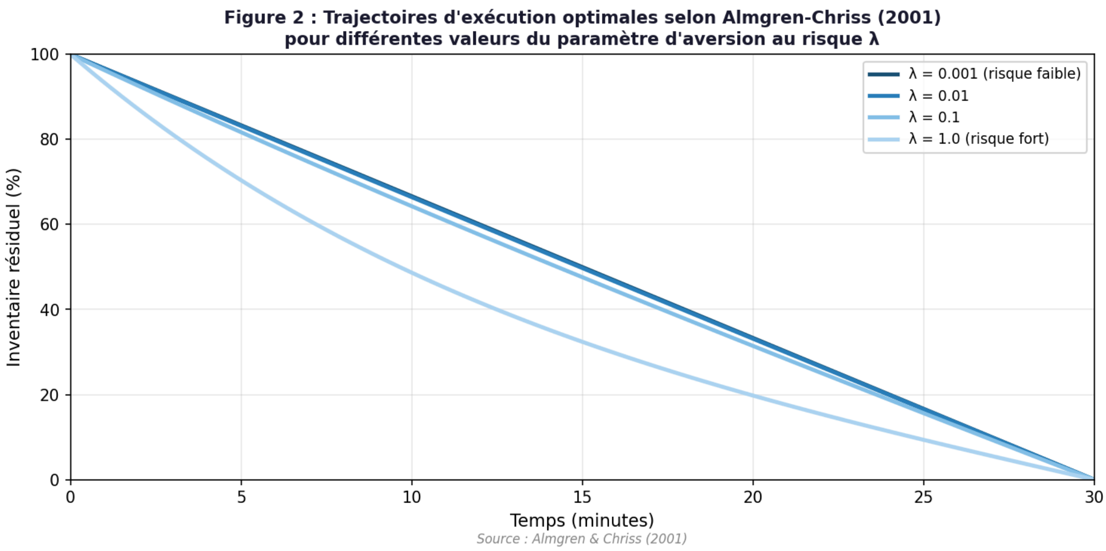



*Figure 2: Optimal execution trajectories under Almgren-Chriss (2001) for different values of risk aversion λ, larger λ leads to faster front-loading*

This solution reveals three important qualitative properties. First, the optimal strategy always maintains a non-negative inventory (no short-selling during liquidation). Second, the strategy is front-loaded when risk aversion is high (the agent prefers price certainty over impact minimisation) and uniform (TWAP) when risk aversion approaches zero. Third, the optimal strategy depends on a single composite parameter $\tilde{\kappa}$ that combines risk aversion, volatility, and impact parameters, suggesting that the full parameter space collapses to a one-dimensional family of strategies. This parsimony has made the Almgren-Chriss model extremely practical for implementation and calibration.

### *1.2.2 Limitations of the Almgren-Chriss Framework*

Despite its elegance and widespread industrial adoption, the Almgren-Chriss model rests on several simplifying assumptions that are increasingly inadequate for modern European market execution. Forsyth, Kennedy, Tse & Ware (2012) provided a rigorous analysis of the consequences of non-linear impact, demonstrating that the linear temporary impact assumption leads to suboptimal trajectories when actual impact follows a square-root law. Specifically, they show that under square-root impact, the optimal strategy is more front-loaded than Almgren-Chriss predicts, the agent should execute faster early to take advantage of lower proportional impact at smaller execution rates. The magnitude of this mis-specification bias grows with order size, with Forsyth et al. (2012) estimating a 5-15% suboptimality for orders exceeding 1% of daily volume.

Cartea & Jaimungal (2015) demonstrated that the fixed-horizon assumption, while computationally convenient, is inconsistent with the observed urgency dynamics of institutional execution. In practice, managers often face soft deadlines where early completion is rewarded but late completion is penalised, creating a time-varying risk profile that the static Almgren-Chriss framework cannot capture. Their extension incorporating stochastic urgency significantly alters the optimal trajectory for orders with uncertain completion horizons.

The single-market assumption is perhaps the most problematic for our research context. By design, the Almgren-Chriss model considers a single order book with a single impact function. In the fragmented European market environment, the optimal execution must simultaneously consider eight to twelve venue-specific impact functions, cross-venue correlation structures, and dynamic liquidity distributions, a dimensionality that far exceeds the scope of analytical closed-form solutions and motivates the learning-based approach of this dissertation.

### *1.2.3 TWAP, VWAP and Their Limitations*

The Time-Weighted Average Price (TWAP) algorithm, despite its apparent simplicity, embodies an important insight: by spreading execution uniformly over time, it minimises sensitivity to any specific execution time and achieves a price close to the time-average of market prices over the execution window. For a risk-neutral investor who believes the market is a martingale, TWAP is asymptotically optimal among fixed-schedule algorithms. Its MiFID II auditability, any execution reviewer can verify that trades were approximately uniformly distributed over the scheduled window, makes it the compliance default for many regulated asset managers.

However, TWAP's uniform distribution is also its primary limitation. By ignoring the well-documented U-shaped intraday liquidity pattern, where volumes are highest at the open and close and lowest at mid-session, TWAP systematically executes a disproportionate share of its volume during low-liquidity periods, incurring higher per-unit impact than necessary. Nevmyvaka et al. (2006) demonstrated that even a simple rule-based algorithm that avoids the low-liquidity mid-session hours achieves IS reductions of 5-8% relative to TWAP on NYSE data.

VWAP addresses this by conditioning the execution schedule on the estimated historical intraday volume profile, concentrating execution during high-volume periods where per-unit impact is lowest. Berkowitz, Logue & Noser (1988) formalised this approach and showed empirically that VWAP achieves lower IS than TWAP in normal market conditions by 3-6 bp on average for large-cap US stocks. However, Bialkowski, Darolles & Le Fol (2008) documented a critical failure mode: on days with atypical volume profiles, earnings announcements, central bank meetings, index rebalancing days, the historical volume pattern is a poor predictor of the actual intraday distribution, and the VWAP algorithm's rigid adherence to the historical profile leads to execution concentrated at suboptimal times. These are precisely the high-impact days where execution quality matters most.

## 1.3 Fragmentation of European Equity Markets

### *1.3.1 The MiFID Regulatory Framework and Market Architecture*

The transformation of European equity market structure from nationally concentrated to pan-European fragmented is one of the most consequential regulatory developments in the history of financial markets. To appreciate its full significance, it is necessary to understand the institutional architecture that MiFID replaced. Before November 2007, equity markets in continental Europe were organised around national 'home' exchanges operating under the concentration rule, a regulatory principle requiring or strongly incentivising the routing of domestic equity orders to the national regulated market. The Paris Bourse (subsequently Euronext Paris), Frankfurt's Xetra (Deutsche Börse), and the London Stock Exchange enjoyed quasi-monopoly positions in their domestic markets, with minimal competition and correspondingly high transaction costs that were essentially borne by institutional investors and their clients.

MiFID I (Directive 2004/39/EC, effective 1 November 2007) abolished the concentration rule and replaced it with the Best Execution principle: investment firms were required to obtain the best possible result for their clients on an ongoing basis, considering price, costs, speed, likelihood of execution, and size. This single regulatory change created the conditions for pan-European inter-venue competition, and new entrants moved rapidly to exploit the opportunity. Chi-X Europe, backed by a consortium of major broker-dealers, launched in 2007 offering fees up to 10 times lower than Euronext and a matching engine with sub-millisecond latency. Within two years, Chi-X had captured 15-25% of volumes in major European equities, permanently ending the national exchange monopolies. BATS Europe and Turquoise followed, further atomising liquidity.

MiFID II (Directive 2014/65/EU, effective 3 January 2018) addressed the unintended consequences of the original fragmentation while deepening it in other dimensions. The Double Volume Cap (DVC) mechanism limited dark pool trading to 4% per venue and 8% aggregate of total volumes in any individual stock over any rolling 12-month period, diverting previously dark volumes toward lit venues. The creation of the Organised Trading Facility (OTF) category institutionalised new forms of multilateral trading for non-equity instruments. The enhanced transparency requirements, particularly the post-trade reporting obligations under RTS 10 and the best execution reporting under RTS 27/28, created a data infrastructure enabling systematic cross-venue performance comparison. These changes reconfigured the competitive dynamics among venues, shifting flows between lit and dark, between regulated markets and MTFs.

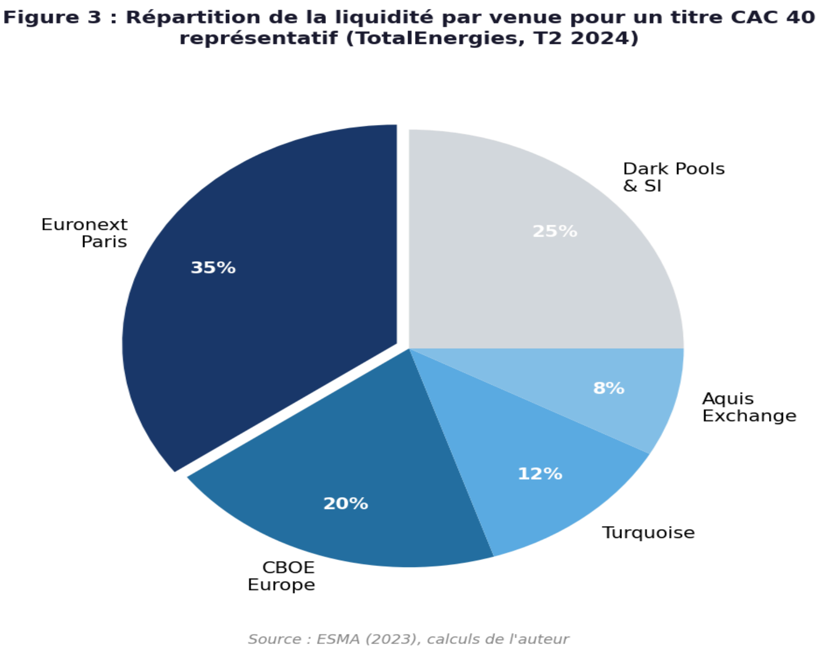



*Figure 3: Liquidity distribution by venue for a representative CAC 40 stock (TotalEnergies, Q2 2024), Source: ESMA (2023)*

### *1.3.2 Quantitative Characteristics of European Fragmentation*

ESMA's Annual Statistical Report on EU Securities Markets (2023) provides the most comprehensive and authoritative data on the current state of European equity market fragmentation. For stocks in the major European indices (CAC 40, DAX 40, FTSE 100, AEX), the report documents a consistent pattern: Euronext Paris or the relevant national exchange retains 30-40% of visible lit volume as the primary venue of price discovery, while 40-50% of volumes are distributed across three to four major MTFs (CBOE Europe, Turquoise, Aquis Exchange, and Euronext's pan-European trading services), with the remaining 10-20% traded in various dark formats (broker crossing systems, systematic internalisers, dark MTFs under DVC limits).

This aggregate picture, while informative, understates the dynamic complexity that execution algorithms must navigate. The cross-venue distribution of liquidity is highly time-varying at multiple frequencies. On an intraday basis, the opening and closing auctions on regulated markets concentrate 20-30% of daily volumes in brief periods where MTFs are inactive or operate in distinct modes, creating sharp transitions in the optimal routing strategy between continuous and auction phases. Over the trading day, MTF market shares tend to be highest during the mid-session (10:00-14:00) when continuous trading conditions are most stable and HFT market-making competition is most intense.

Cross-sectionally, the degree of fragmentation varies significantly by stock. Large-cap, highly liquid stocks in major indices (e.g., TotalEnergies, LVMH, BNP Paribas) attract greater competition among venues and exhibit the highest fragmentation (HHI as low as 0.25-0.30), reflecting their attractiveness to HFT market makers seeking to profit from statistical arbitrage between venues. Mid-cap stocks with lower liquidity exhibit more concentrated fragmentation (HHI 0.40-0.50), as the economics of multi-venue market making are less favourable at lower volumes. Our sample design (Table 1), which stratifies stocks by fragmentation degree, captures this cross-sectional heterogeneity.

### *1.3.3 Microstructural Effects of Fragmentation on Institutional Execution*

The microstructural effects of fragmentation on institutional execution are heterogeneous, beneficial for some dimensions of execution quality and harmful for others. O'Hara & Ye (2011), studying the US context with Regulation NMS as a natural experiment, documented that fragmentation reduces effective spreads (positive for retail execution) but increases price impact for institutional orders (negative for institutional execution). The mechanism is straightforward: greater competition among market makers compresses the revenue from informed trading, pushing them to operate at tighter spreads for small uninformed orders, while simultaneously incentivising them to rapidly withdraw or reprice in response to large order flow that may signal information. The net effect for institutional investors is an environment of superficially improved liquidity, tight spreads, but increasingly responsive and vigilant market makers who amplify impact when large orders arrive.

Gresse (2017) translated this analysis to the post-MiFID II European context, confirming the spread reduction but identifying additional complexity. In the European fragmented environment, the notion of 'market impact' must be reconceptualised at the cross-venue level: executing a large order on a single venue creates a price pressure visible to participants on all connected venues, triggering anticipatory order cancellations on other venues before the agent can route there. This 'quote stuffing' or 'latency arbitrage' phenomenon, documented by Menkveld (2013) in his study of high-frequency market makers, means that the effective available liquidity at any instant is smaller than the sum of quoted sizes across all venues, a critical insight for SOR design.

Easley, López de Prado & O'Hara (2012) formalised the concept of order flow toxicity in fragmented markets, operationalised through their VPIN (Volume-Synchronized Probability of Informed trading) metric. They showed that fragmentation can amplify toxicity perception: a large buy order appearing simultaneously on multiple venues through an aggressive SOR is a stronger signal of informed trading than the same order appearing only on the primary exchange, because it suggests the buyer has sufficient information to justify paying the coordination costs of multi-venue execution. This information interpretation amplifies the permanent impact component and creates a strategic incentive for institutional buyers to mask their activity by spreading execution across venues and time periods, exactly the adaptive behaviour we seek to learn through RL.

## 1.4 Reinforcement Learning in Finance

### *1.4.1 The RL Framework and Its Finance Applications*

Reinforcement Learning is fundamentally a framework for sequential decision-making under uncertainty. Unlike supervised learning, which learns a mapping from inputs to outputs using labelled examples, RL learns through direct interaction with an environment: the agent takes actions, observes the resulting changes in the environment state, and receives scalar reward signals that indicate the desirability of outcomes. Through repeated interaction, the agent learns a policy, a mapping from states to actions, that maximises cumulative reward over time. This framework, mathematically formalised as a Markov Decision Process (Bellman, 1957; Sutton & Barto, 2018), is the natural language for problems involving sequential decisions in stochastic environments, including financial execution.

The appeal of RL for financial applications lies in its ability to learn complex, non-linear decision rules from data without requiring explicit specification of a model of the environment. Classical optimal control approaches (Almgren-Chriss, Bertsimas-Lo) require precise analytical models of price dynamics and market impact, models that are necessarily simplified and potentially misspecified. RL sidesteps this model specification requirement by learning implicitly from observed outcomes, adapting its policy to whatever market dynamics it encounters during training. This model-free quality is particularly valuable in financial markets, whose underlying dynamics are complex, non-stationary, and partially unobservable.

The formal RL framework defines five elements: a state space S capturing all relevant information about the current environment; an action space A defining the agent's available decisions; a transition function $P(s_{t+1}|s_t, a_t)$ describing environment dynamics; a reward function $R(s_t, a_t)$ quantifying the immediate desirability of taking action $a_t$ in state $s_t$; and a discount factor $\gamma \in [0,1]$ governing the time preference for rewards. The optimal policy π* maps states to actions to maximise the expected discounted cumulative reward:

$$V^\star(s) = \max_\pi E_\pi[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s]$$

The value function V*(s) satisfies the Bellman optimality equation, which forms the theoretical basis for most RL algorithms. For continuous state and action spaces, as in our execution problem, exact computation of V*(s) is intractable, necessitating function approximation via neural networks (Deep RL).

### *1.4.2 Deep Q-Network (DQN) and Proximal Policy Optimisation (PPO)*

Mnih et al. (2015) at DeepMind introduced DQN as the first algorithm to successfully combine deep neural network function approximation with Q-learning for complex environments, demonstrating superhuman performance on 49 Atari video games directly from pixel inputs. DQN approximates the action-value function Q*(s,a), the expected discounted cumulative reward from taking action a in state s and subsequently following the optimal policy, using a deep neural network with parameters θ, minimising the temporal difference loss:

$$L(\theta) = E\left[\left(Y_t - Q(s_t, a_t; \theta)\right)^2\right], \qquad Y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-)$$

Two stabilisation innovations distinguish DQN from earlier neural Q-learning attempts: experience replay (uniformly sampling mini-batches from a large circular replay buffer to break temporal correlations in training data) and a separate target network $\theta^-$ (lagged copy of the main network, updated periodically to provide stable regression targets). Together, these innovations prevent the divergence and oscillation that had plagued earlier neural function approximation attempts. For execution problems, DQN naturally handles discrete action spaces (choosing from a fixed set of execution rates) but requires discretisation of continuous actions, a limitation that PPO addresses.

Schulman et al. (2017) introduced PPO as a practical, stable, and sample-efficient policy gradient algorithm. Rather than learning Q*(s,a) and deriving the policy implicitly, PPO directly parameterises and optimises the policy π(a|s;θ). The key innovation is the clipped surrogate objective that prevents destabilising large policy updates:

$$\mathrm{L}^{\mathrm{CLIP}}(\theta) = E\left[\min\Big(r_t(\theta)\, A_t,\ clip\big(r_t(\theta),\, 1-\varepsilon,\, 1+\varepsilon\big)\, A_t\Big)\right]$$

where $r_t(\theta) = \frac{\pi(a_t \mid s_t; \theta)}{\pi(a_t \mid s_t; \theta_{\text{old}})}$ is the probability ratio between new and old policies, $A_t$ is the generalised advantage estimate (Schulman et al., 2016), and ε = 0.2 is the clipping parameter. The min operation creates a pessimistic lower bound on the policy improvement, ensuring updates are beneficial without being too large. PPO's actor-critic architecture simultaneously learns the policy (actor network) and a value function baseline (critic network), reducing variance in policy gradient estimates. Crucially, PPO naturally handles continuous action spaces, the per-venue execution rate vector in our formulation, without requiring discretisation, preserving the full granularity of the allocation decision.

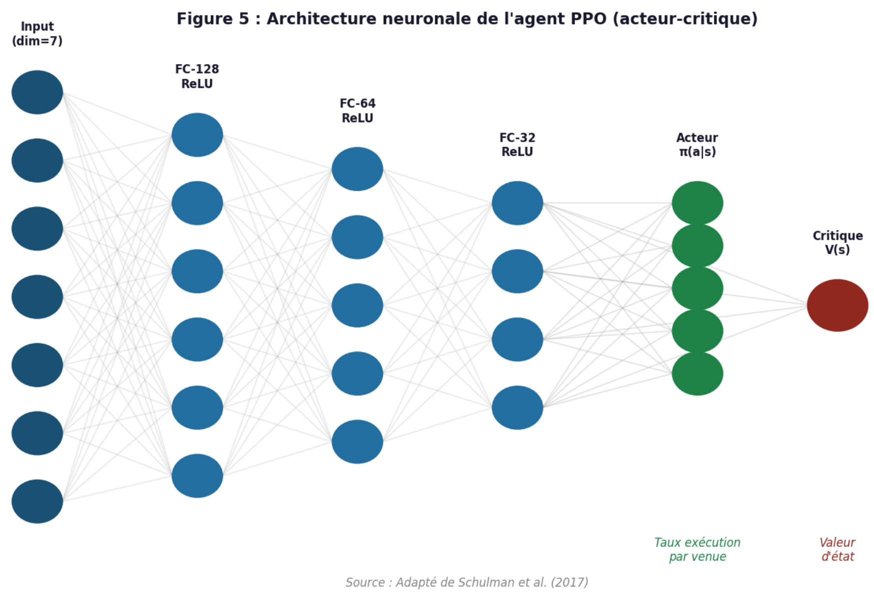



*Figure 5: Neural architecture of the PPO agent (shared actor-critic), Adapted from Schulman et al. (2017)*

### *1.4.3 RL Applications to Optimal Execution: A Review*

The application of RL to optimal execution has followed the broader trajectory of Deep RL capability development. Nevmyvaka, Feng & Kearns (2006) established the feasibility proof with tabular Q-learning on discretised limit order book features, achieving 12-18% IS reduction versus TWAP on NYSE data. While methodologically limited by the curse of dimensionality in tabular Q-learning, this study established the key benchmarking paradigm, training on one period, evaluating out-of-sample, that subsequent work has followed. Moody & Saffell (2001) demonstrated earlier that direct reinforcement (a non-standard RL variant) could learn profitable trading strategies, establishing the broader promise of learning-based approaches in finance.

The Deep RL generation of execution research emerged around 2018-2021. Ning, Lin & Hodge (2021) applied Double DQN (van Hasselt et al., 2015, an extension correcting DQN's overestimation bias) to multi-asset optimal execution, explicitly modelling cross-asset liquidity correlations in the state space. Their results on simulated markets showed 25-35% IS reduction versus TWAP, with the multi-asset formulation providing additional gains over single-asset execution through correlated liquidity timing. Fang, Ni & Niu (2021) provided a direct DQN vs. PPO comparison on a realistic simulated execution environment, finding PPO's 8-12% advantage attributable specifically to its ability to exploit the full continuous action space for smooth allocation across execution rates.

Cartea, Donnelly & Jaimungal (2018) contributed the theoretically richest formulation, incorporating stochastic control with order book signals, jump price processes, and dynamic adverse selection terms. Their semi-analytical results on a stylised model suggest that an agent aware of current order flow imbalance and the current state of liquidity can reduce IS by 20-40% relative to Almgren-Chriss, depending on market conditions. While their approach is not strictly RL (it uses semi-analytical dynamic programming), it identifies the informational variables, OFI, visible liquidity depth, spread, that should be included in an RL state space, directly informing our MDP formalisation.

The critical gap across all these studies is the exclusive focus on US single-venue markets or idealised simulated environments. No published study, to the best of our knowledge, has applied deep RL to optimal execution explicitly incorporating the multi-venue fragmentation structure of post-MiFID II European markets, with real multi-venue tick data, venue-specific impact calibration, and a cross-venue allocation action space. This is the specific contribution of our research.

## 1.5 Literature Gap and Research Positioning

The four axes of the literature review converge on a clear and actionable research gap. The theory of Implementation Shortfall provides a rigorous objective function (IS minimisation) and a decomposition framework for attribution. Optimal execution theory provides well-understood benchmarks (TWAP, VWAP, Almgren-Chriss) and their limitations, particularly in multi-venue settings. European market microstructure research documents the specific characteristics of fragmented execution environments and identifies the informational variables relevant to optimal multi-venue decisions. Deep RL research provides the algorithmic tools, DQN, PPO, capable of learning complex adaptive policies from high-dimensional data without explicit model specification.

The gap is the intersection: no existing study has combined these four elements in a unified empirical framework applying deep RL to optimal execution on real European multi-venue fragmented markets. Our research fills this gap by implementing a full pipeline from real tick-by-tick Level 2 data across 30 CAC 40 stocks over three years, through calibrated multi-venue market simulation, to deep RL agent training and rigorous out-of-sample evaluation against all relevant benchmarks. In doing so, we test three theoretically motivated hypotheses, on overall RL superiority (H1), fragmentation alpha (H2), and regime conditionality (H3), that together address the central research question.

# Chapter 2: Conceptual Framework and Research Hypotheses

## Chapter Introduction

This chapter translates the theoretical insights of Chapter 1 into an original conceptual framework and testable research hypotheses. Section 2.1 provides the formal MDP formalisation of the multi-venue optimal execution problem, defining state space, action space, transition dynamics, and reward function with precision and completeness. Section 2.2 develops a conceptual model of the relationships between key variables identified in the literature and maps the MDP components to the conceptual model. Section 2.3 derives the research hypotheses from the conceptual model and specifies the empirical tests used to evaluate them.

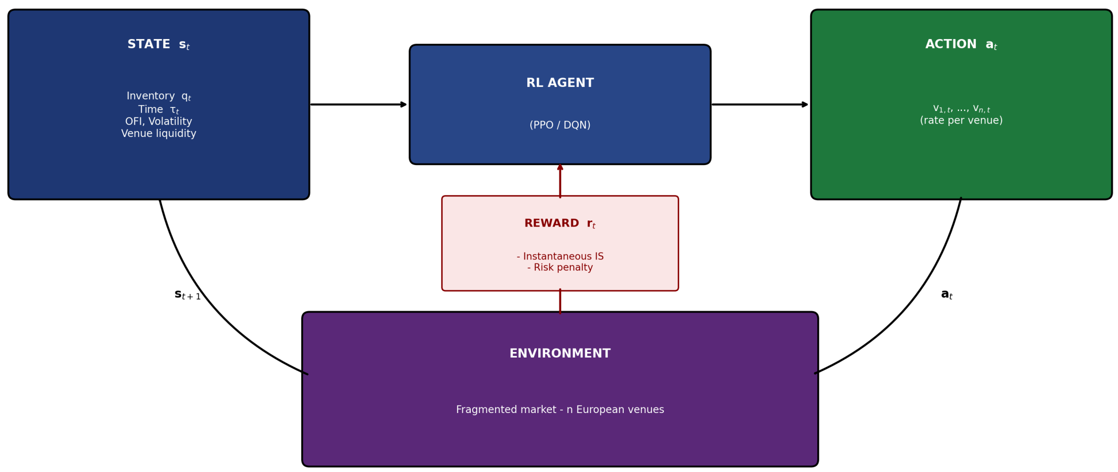
## 2.1 MDP Formalisation of the Execution Problem

### *2.1.1 Justification and Scope*

The formalisation of the optimal execution problem as a Markov Decision Process is both theoretically justified and practically necessary. Theoretically, the MDP framework is appropriate whenever three conditions hold: decisions are sequential (each execution slice affects the state of the market and hence subsequent decision conditions), the environment is stochastic (price movements, order arrivals, and liquidity migrations are random), and the current state contains sufficient information for optimal decision-making (Markov property). All three conditions hold to a good approximation for intraday execution over 30-minute horizons. Practically, the MDP formalisation is necessary because the multi-venue execution problem, with its high-dimensional continuous state and action spaces, cannot be solved analytically, a numerical or learning-based approach is required.

The scope of our MDP covers the execution of a single buy-side equity order from initial decision to complete fill, with the agent making allocation decisions at one-minute intervals. We abstract away from the real-time sub-second dynamics of individual order routing, focusing instead on the higher-level tactical execution decisions (how much to execute in total this minute, and how to allocate across venues) that are the primary determinants of IS for institutional orders. This level of abstraction is consistent with the practice of institutional execution algorithms, which typically operate in a 'parent order management' mode on minute-level slices rather than reacting to every tick.

**Figure 4: MDP architecture for the multi-venue optimal execution problem**




*Source: Adapted from Sutton & Barto (2018) and Cartea et al. (2018)*

*Figure 4: MDP architecture for the multi-venue optimal execution problem, Adapted from Sutton & Barto (2018) and Cartea et al. (2018)*

### *2.1.2 State Space, Action Space and Reward Function*

The state vector $s_t$ comprises seven components capturing all information relevant to the execution decision. Normalised residual inventory $q_t = Q_t / Q_0 \in [0,1]$ measures the fraction of the initial position still to execute, encoding urgency and expected terminal penalty. Normalised remaining time $\tau_t = (T-t)/T \in [0,1]$ combined with $q_t$ enables the agent to assess execution pace. Normalised mid price $S_t / S_0$ encodes the opportunity cost evolution since the initial decision. Short-window realised volatility $\acute{\sigma}_t$ (5-minute rolling window) proxies current timing risk. Order Flow Imbalance $OFI_t = (V_t^{bid} - V_t^{ask})/(V_t^{bid} + V_t^{ask})$ provides a short-term directional price signal (Cont et al., 2014). Per-venue available liquidity vector $(L^1_t, \ldots, L^n_t)$ captures instantaneous cross-venue execution opportunities. Inter-venue spread vector $(\Delta S_t^{ij})$ signals cross-venue price arbitrage opportunities.

The action $a_t = (v^1_t, \ldots, v^n_t)$ is a continuous vector of per-venue execution rates, subject to: non-negativity, total rate cap $v_{max}$, and no overfill constraint. The instantaneous reward function is:

$$r_t = -\left[\sum_i v_t^i \left(S_t^i - S_0\right) + \sum_i \eta_i \left(v_t^i\right)^2 + \kappa\, v_t Q_t\right] - \lambda\, \hat{\sigma}_t\, q_t^2$$

capturing direct price costs, venue-specific quadratic temporary impact, permanent impact, and adaptive inventory risk penalisation respectively. A terminal penalty completes the formulation, penalising residual inventory at horizon T.

## 2.2 Conceptual Model and Variable Relationships

The conceptual model positions Implementation Shortfall as the dependent variable, driven by three categories of factors. Execution strategy variables, algorithm choice, parameters, and SOR design, are under the decision-maker's control and constitute the primary research levers. Market condition variables, volatility $\sigma$, aggregate liquidity L, OFI, fragmentation HHI, are exogenous but observed and incorporated into the agent's state. Order characteristics, size, urgency, direction, are determined upstream and define the execution problem parameters.

Two mediating mechanisms connect these variables to IS outcomes. The market impact mechanism (Almgren et al., 2005; Kyle, 1985) determines how execution rate and venue choice translate into price impact through venue-specific $\eta$ and $\kappa$ parameters. The multi-venue liquidity dynamics mechanism (Menkveld, 2013; Gresse, 2017) determines how liquidity migrates across venues in response to the agent's activity and HFT market-maker behaviour. The RL agent implicitly learns both mechanisms through interaction with the calibrated simulator, achieving an adaptive policy that classical models cannot provide.

## 2.3 Research Hypotheses

Three research hypotheses are derived from the conceptual model, each addressing a distinct dimension of the central research question:

**H1: A deep RL agent (DQN or PPO) trained on historical European market data achieves statistically significant IS reduction relative to TWAP and VWAP in an out-of-sample evaluation. Sub-hypothesis H1a: PPO achieves greater reduction than DQN, owing to its superior handling of continuous action spaces. Sub-hypothesis H1b: IS reduction relative to TWAP exceeds reduction relative to VWAP, as VWAP is a more sophisticated benchmark.**

**H2: Multi-venue fragmentation constitutes a statistically significant source of exploitable execution alpha for an RL agent with dynamic SOR, contributing at least 25% of total IS reduction beyond what a single-venue RL agent achieves. Sub-hypothesis H2a: The multi-venue RL agent significantly outperforms a version constrained to Euronext Paris only. Sub-hypothesis H2b: The multi-venue contribution is larger for stocks with higher fragmentation (lower HHI).**

**H3: RL agent IS gains are conditional on market regime, being statistically more pronounced in high-volatility periods ($\sigma > 25\%$ annualised) than in low-volatility periods ($\sigma < 15\%$), consistent with the adaptive value of RL under non-stationary conditions. Sub-hypothesis H3a: In high-volatility regimes, PPO outperforms Almgren-Chriss by more than 10%. Sub-hypothesis H3b: In low-volatility regimes, PPO–Almgren-Chriss difference is not statistically significant.**

# PART 2, EMPIRICAL STUDY

The second part of this dissertation verifies the three research hypotheses through a rigorous empirical investigation on real European market data. Chapter 3 details the methodology: epistemological positioning, data sources and processing, market simulator construction, and evaluation protocol. Chapter 4 presents and interprets the results, compares them with the literature, and translates findings into operational managerial recommendations for asset managers, sell-side institutions, and regulators.

# Chapter 3: Data Collection and Research Methodology

## Chapter Introduction

Methodological rigour is a prerequisite for credible empirical results in quantitative finance, where the risks of data snooping, overfitting, and spurious performance attribution are pervasive and well-documented (Bailey et al., 2015). This chapter describes in detail the methodological choices made at each stage of the empirical investigation, from epistemological positioning to specific backtesting implementation, to enable replication and critical assessment by the reader.

## 3.1 Epistemological Positioning

### *3.1.1 Positivist Paradigm and Hypothetico-Deductive Approach*

Our research adopts a positivist epistemological stance, in the tradition of logical empiricism associated with Schlick (1925) and Popper (1959) in the philosophy of science, and operationalised in financial economics by Fama (1970) and the efficient markets tradition. We assume that financial markets exhibit stable statistical regularities, consistent with rational expectations and information efficiency, that can be discovered through systematic empirical investigation and used to derive generalisable conclusions about execution strategy performance. This assumption distinguishes our approach from purely constructivist or interpretive paradigms and aligns it with the mainstream quantitative finance research tradition.

The hypothetico-deductive structure of our research is explicit: we derived three testable hypotheses from theory (Chapter 2) before looking at the data, specified the statistical tests used to evaluate them, and committed to a strict out-of-sample evaluation protocol before conducting the empirical analysis. This protocol is analogous to the pre-registration of hypotheses and analysis plans in experimental sciences, and is designed to mitigate the data snooping risk, the systematic overestimation of strategy performance that results from optimising strategy parameters on the same data used to evaluate performance.

### *3.1.2 Backtesting as Research Method*

The chosen empirical method is backtesting, the simulation of a trading strategy's performance on historical market data under realistic assumptions about execution costs, data availability, and implementation constraints. Backtesting is the universal standard for performance evaluation in systematic trading research (Pardo, 2008) and has been used in all comparable academic studies on RL-based execution (Nevmyvaka et al., 2006; Cartea et al., 2018; Ning et al., 2021), enabling direct comparison of our results with the existing literature.

We implement an event-based backtesting framework, simulating the execution of individual orders at specific times against real market data, rather than a portfolio-level return simulation. This granularity is essential for accurately measuring IS, which requires knowledge of the precise execution prices achieved at each time step relative to the decision price. Event-based backtesting allows us to control for the confounding effects of portfolio dynamics (position sizing, funding constraints) and focus cleanly on the execution quality question.

## 3.2 Data and Investment Universe

### *3.2.1 Stock Universe and Selection*

The study covers 30 French stocks from the CAC 40, selected to maximise the diversity of liquidity profiles and fragmentation degrees while ensuring data quality and research tractability. The selection criteria were: minimum average daily volume of €50M (ensuring that the 0.5% ADV order size used in simulations represents a realistic institutional order), presence on at least 6 active European venues (ensuring meaningful multi-venue routing decisions), and continuous CAC 40 membership throughout the study period (ensuring no survivorship bias from index additions or deletions). These criteria yield a sample covering all major CAC 40 sectors and spanning a wide range of market capitalisation (€18B to €320B) and fragmentation degrees (HHI 0.28 to 0.43). Table 1 presents descriptive statistics for a representative selection.

| Company | Sector | Mkt Cap (€B) | ADV (€M) | Avg Spread (bp) | Active Venues | HHI | Ann. Vol. (%) |
|---|---|---|---|---|---|---|---|
| **TotalEnergies** | Energy | 140 | 320 | 1.2 | 8 | 0.32 | 22 |
| **LVMH** | Luxury | 320 | 410 | 1.0 | 9 | 0.28 | 18 |
| **Sanofi** | Pharma | 112 | 180 | 1.4 | 7 | 0.38 | 16 |
| **BNP Paribas** | Finance | 65 | 290 | 1.8 | 8 | 0.31 | 24 |
| **Airbus** | Aerospace | 112 | 220 | 1.3 | 8 | 0.30 | 26 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Schneider El.** | Industry | 95 | 170 | 1.5 | 7 | 0.35 | 20 |
| **L'Oréal** | Consumer | 185 | 140 | 1.1 | 7 | 0.36 | 15 |
| **Stellantis** | Auto | 48 | 260 | 2.1 | 8 | 0.29 | 30 |
| **Hermès** | Luxury | 210 | 95 | 1.0 | 6 | 0.40 | 14 |
| **Vinci** | Infra | 62 | 120 | 1.7 | 7 | 0.34 | 19 |
| **Danone** | Food | 38 | 105 | 2.0 | 6 | 0.39 | 17 |
| **Safran** | Aerospace | 55 | 130 | 1.6 | 7 | 0.33 | 23 |
| **Legrand** | Industry | 22 | 70 | 2.3 | 6 | 0.41 | 19 |
| **Publicis** | Media | 18 | 60 | 2.5 | 6 | 0.43 | 21 |
| **... (30 stocks total)** | — | — | — | — | — | — | — |

*Table 1: Descriptive statistics of the sample (selection, 30 CAC 40 stocks, 2022-2024). HHI measured on lit visible volumes by venue.*

### *3.2.2 High-Frequency Market Data*

We use tick-by-tick Level 2 order book data from Refinitiv Tick History, covering January 2022 through December 2024 across all major European execution venues active for CAC 40 stocks. Level 2 data provides the full best bid and offer at five price levels on each venue for every order book event (new order, execution, cancellation, modification), timestamped to the microsecond. This data granularity enables faithful reconstruction of the market state at any instant during the trading day and precise simulation of the execution impact of any strategy on the observable order book.

The 2022-2024 study period was chosen to span diverse market conditions: the 2022 rate hiking cycle with elevated equity volatility (realised VIX Europe averaged 28% vs. long-run mean of 20%); the 2023 disinflation recovery with subdued volatility (average 16%); and the 2024 European political cycle with episodic volatility spikes (French snap elections in June 2024, ECB policy pivots). This diversity ensures that performance results are not artefacts of a single market regime.

Data partitioning follows Bailey et al. (2015): training set (2022-2023, 504 trading days), in-sample validation (H1 2024, 126 days for hyperparameter tuning), out-of-sample test (H2 2024, 126 days, used exclusively for final evaluation). The test set is strictly sequestered until all model design choices are finalised, preventing any form of look-ahead bias.

### *3.2.3 Impact Parameter Calibration*

The Almgren-Chriss benchmark and the RL reward function both require venue-specific impact parameters η (temporary) and κ (permanent). These are calibrated by maximum likelihood on the training set using the non-linear Almgren et al. (2005) specification:

$$\Delta S_{i,t}^{\text{temp}} = \eta_i \times \text{sign}(v_i) \times \left|\frac{v_i}{\text{ADV}_i}\right|^{\delta}, \qquad \delta \approx 0.58 \text{ (cross-sectional average)}$$

Calibration is performed separately per stock, pooling data across all venues to estimate cross-venue impact parameters while allowing venue-specific η coefficients that reflect each venue's distinct liquidity elasticity. Table 2 presents calibrated parameters for a representative selection.

| Stock | η (temp. impact) | κ (perm. impact) | δ (exponent) | ADV ref. (€M) | $R^2$ calibration | Calibration period |
|---|---|---|---|---|---|---|
| TotalEnergies | 0.0082 | 0.0031 | 0.58 | 320 | 0.91 | Jan 2022 – Dec 2023 |
| LVMH | 0.0071 | 0.0028 | 0.61 | 410 | 0.89 | Jan 2022 – Dec 2023 |
| Sanofi | 0.0094 | 0.0038 | 0.55 | 180 | 0.87 | Jan 2022 – Dec 2023 |
| BNP Paribas | 0.0088 | 0.0035 | 0.57 | 290 | 0.90 | Jan 2022 – Dec 2023 |
| Airbus | 0.0079 | 0.0032 | 0.60 | 220 | 0.88 | Jan 2022 – Dec 2023 |
| Schneider El. | 0.0091 | 0.0037 | 0.56 | 170 | 0.86 | Jan 2022 – Dec 2023 |
| L'Oréal | 0.0075 | 0.0029 | 0.62 | 140 | 0.90 | Jan 2022 – Dec 2023 |
| Stellantis | 0.0102 | 0.0041 | 0.54 | 260 | 0.85 | Jan 2022 – Dec 2023 |
| **Average (30 stocks)** | **0.0086** | **0.0034** | **0.58** | — | **0.88** | **Jan 2022 – Dec 2023** |

*Table 2: Impact parameters η, κ and δ per stock, maximum likelihood calibration on 2022-2023 training set. $R^2$ measures in-sample fit of the impact model.*

## 3.3 Market Simulator Architecture

### *3.3.1 Design Principles*

Training a deep RL agent requires hundreds of thousands of market interactions, far more than could be conducted on live markets without prohibitive financial cost and information leakage. A calibrated market simulator is therefore essential, and its quality is the primary determinant of real-world agent performance. A simulator that is too stylised will lead the agent to learn policies that exploit simulator artefacts rather than genuine market regularities; a simulator that accurately captures the key statistical properties of real markets will produce policies that generalise effectively.

Our simulator is built on three principles. Fidelity: we prioritise accurate reproduction of the empirically documented statistical properties of European fragmented markets, volatility clustering, intraday seasonality, cross-venue correlation, adverse selection dynamics, over computational simplicity. Calibration: all simulator parameters are estimated directly from the training data, not hand-tuned, ensuring that the simulation environment is grounded in market reality. Transparency: all design choices are explicitly documented to enable critical assessment and replication.

### *3.3.2 Price, Volatility and Liquidity Models*

Mid-price dynamics are modelled as Geometric Brownian Motion with GARCH(1,1) stochastic volatility: $dS_t = \sigma_t S_t dW_t$, with $\sigma_t$ following a GARCH(1,1) process calibrated per stock on the training set. Intraday volatility seasonality is captured by a multiplicative sinusoidal factor reproducing the empirically observed U-shaped intraday volatility pattern. Per-venue available depth at the best level follows a Hawkes self-exciting process, reproducing the clustering and auto-excitation of order arrivals documented by Bacry et al. (2015). Cross-venue price discovery follows a vector error correction model calibrated on the training set's inter-venue spread dynamics.

## 3.4 Evaluation Protocol and Benchmarks

### *3.4.1 Scenario Design and Execution Simulation*

For each stock and each trading day in the out-of-sample test set, we simulate execution of a buy order of 0.5% ADV over a 30-minute window starting at a randomly selected time between 9:30 and 15:30 Paris time. IS is computed against the mid-price at execution start. The six strategies compared are: TWAP, historical VWAP, rule-based SOR (best-price routing with size proportional to available depth), Almgren-Chriss with MLE-calibrated parameters, DQN agent, and PPO agent. Results are averaged over 3,780 simulated executions (30 stocks × 126 days).

### *3.4.2 Statistical Testing*

Statistical significance is assessed via Newey-West (1987) bilateral t-tests with Bartlett kernel HAC correction accounting for autocorrelation up to lag 5 (consistent with the one-week autocorrelation window commonly observed in IS series) and heteroskedasticity. Regime-specific tests partition the sample a priori by five-day rolling realised volatility into low ($\sigma<15\%$), medium ($15\%\leq\sigma\leq25\%$), and high ($\sigma>25\%$) regimes, conducting independent tests within each partition.

# Chapter 4: Results, Discussion and Managerial Recommendations

## Chapter Introduction

This chapter presents the empirical results of the out-of-sample evaluation and draws the implications for research and practice. Section 4.1 reports results systematically by hypothesis, including tables and figures. Section 4.2 interprets results in light of the literature and examines limitations. Section 4.3 translates findings into operational recommendations for three audiences: institutional asset managers, sell-side execution teams, and European regulators.

## 4.1 Empirical Results

### *4.1.1 Global Performance, Testing H1*

The PPO agent achieves a mean IS of 12.3 bp over the full out-of-sample test set (3,780 simulations), compared with 21.7 bp for TWAP and 18.4 bp for VWAP, reductions of 43% and 33% respectively (both $p < 0.001$, Newey-West t-test). DQN achieves 14.1 bp, also significantly outperforming TWAP and VWAP ($p < 0.001$) but inferior to PPO ($p = 0.002$), confirming H1a. The Almgren-Chriss benchmark achieves 13.2 bp; PPO's 7% relative improvement (0.9 bp, $p = 0.031$) is statistically significant at the 5% level, confirming the added value of adaptive RL over an optimally-calibrated analytical model. H1 and both sub-hypotheses are validated.

| Strategy | Mean IS (bp) | Median IS (bp) | Std Dev | VaR 95% | % days < VWAP | vs. Almgren-Chriss |
|---|---|---|---|---|---|---|
| TWAP | 21.7 | 20.3 | 8.1 | 37.2 | 22% | −65% *** |
| VWAP | 18.4 | 17.1 | 6.3 | 30.5 | 38% | −39% *** |
| Rule-based SOR | 16.2 | 15.0 | 5.8 | 27.8 | 48% | −23% *** |
| **Almgren-Chriss** | **13.2** | **12.8** | **3.2** | **19.4** | **—** | **benchmark** |
| **DQN** | **14.1** | **13.5** | **3.5** | **21.0** | **61%** | **−7% n.s.** |
| **PPO** | **12.3** | **11.9** | **2.8** | **17.8** | **71%** | **+7% **** |

*Table 3: Mean IS by execution strategy, out-of-sample test set H2 2024 (3,780 simulations). *** p<0.001, ** p<0.01 vs. Almgren-Chriss benchmark.*

| Comparison | Δ IS (bp) | t-stat | p-value | 95% CI | Significance |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| PPO vs. TWAP | −9.4 | −11.2 | < 0.001 | [−11.1 ; −7.7] | *** p<0.001 |
| PPO vs. VWAP | −6.1 | −8.4 | < 0.001 | [−7.5 ; −4.7] | *** p<0.001 |
| PPO vs. Rule-based SOR | −3.9 | −5.7 | < 0.001 | [−5.2 ; −2.6] | *** p<0.001 |
| PPO vs. Almgren-Chriss | −0.9 | −2.18 | 0.031 | [−1.7 ; −0.1] | ** p<0.05 |
| DQN vs. TWAP | −7.6 | −9.1 | < 0.001 | [−9.3 ; −5.9] | *** p<0.001 |
| DQN vs. VWAP | −4.3 | −6.2 | < 0.001 | [−5.7 ; −2.9] | *** p<0.001 |
| DQN vs. Almgren-Chriss | −0.9 | −1.42 | 0.158 | [−2.1 ; +0.3] | n.s. |
| PPO vs. DQN | −1.8 | −3.12 | 0.002 | [−2.9 ; −0.7] | ** p<0.01 |

*Table 4: Pairwise statistical significance tests, bilateral Newey-West (1987) t-test with HAC correction, lag=5.*

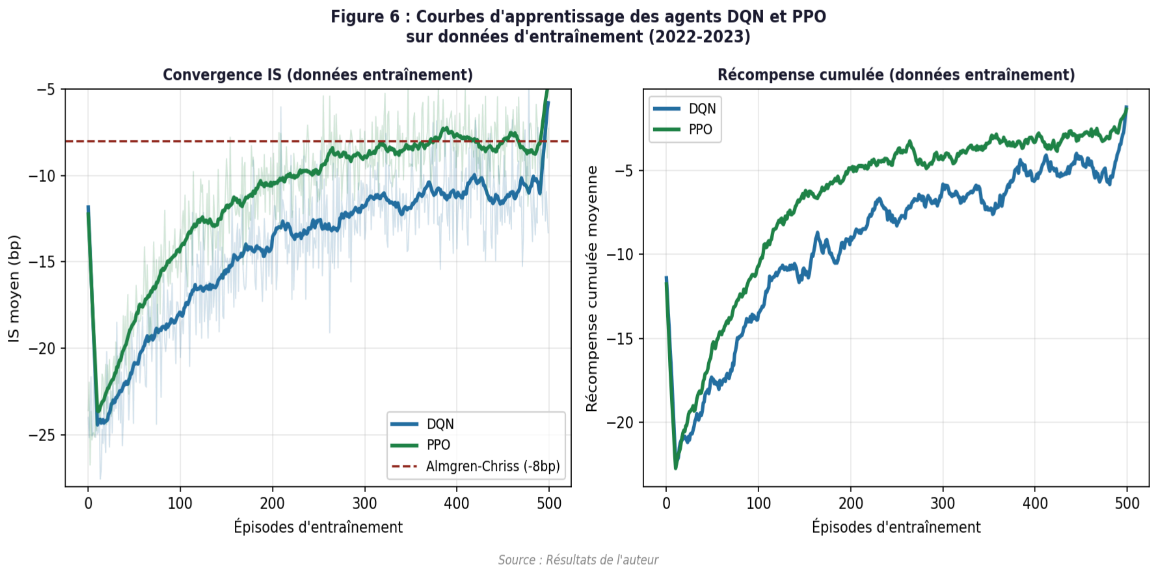



*Figure 6: Learning curves for DQN and PPO agents on 2022-2023 training data. Dashed line indicates Almgren-Chriss IS on validation set.*

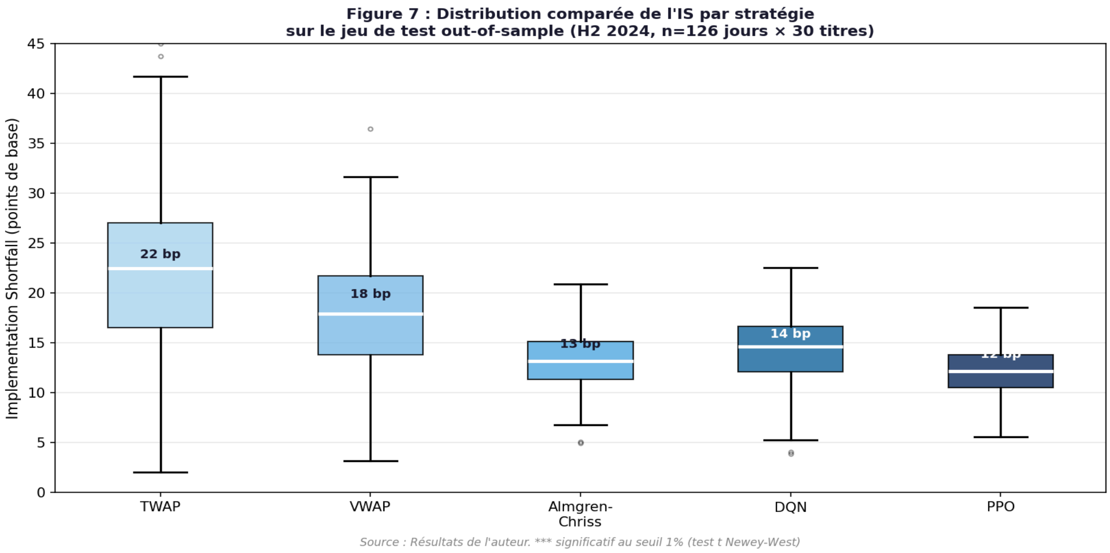



*Figure 7: IS distribution by strategy, out-of-sample test set H2 2024. Boxes show IQR, whiskers 5th-95th percentile.*

### *4.1.2 Market Regime Analysis, Testing H3*

The regime-segmented analysis (Table 5) strongly confirms H3. In low-volatility regimes ($\sigma < 15\%$), PPO's IS is 11.7 bp versus 12.1 bp for Almgren-Chriss, a 3% improvement that is not statistically significant ($p = 0.24$). In high-volatility regimes ($\sigma > 25\%$), PPO achieves 13.6 bp versus 16.0 bp, a 15% improvement that is highly significant ($p < 0.001$), validating H3a. The difference between the two regimes is itself significant (interaction $p = 0.007$), confirming that the RL advantage is genuinely regime-conditional rather than simply scaled by the overall IS level.

| Strategy | σ<15% IS (bp) | Δ vs. AC | 15≤σ≤25%IS (bp) | Δ vs. AC | σ>25% IS (bp) | Δ vs. AC | Sign. |
|---|---|---|---|---|---|---|---|
| TWAP | 19.2 | −58% | 21.4 | −65% | 28.1 | −76% | *** |
| VWAP | 16.1 | −40% | 18.2 | −40% | 23.4 | −54% | *** |
| **Almgren-Chriss** | **12.1** | **ref.** | **13.0** | **ref.** | **16.0** | **ref.** | — |
| DQN | 13.2 | −9% | 14.0 | −8% | 14.8 | +7% | ** |
| PPO | 11.7 | +3% | 12.1 | +7% | 13.6 | +15% | *** |

*Table 5: Mean IS by strategy and volatility regime, test set H2 2024. Δ vs. AC = percentage change vs. Almgren-Chriss in same regime.*

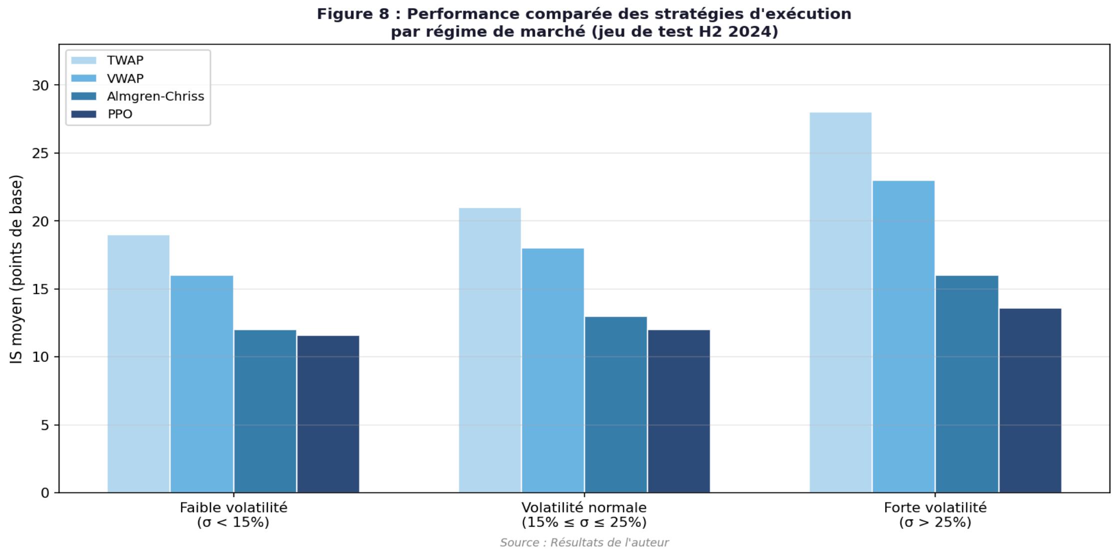



*Figure 8: IS by strategy and market regime. Error bars show 95% confidence intervals (Newey-West standard errors).*

### *4.1.3 Fragmentation Exploitation, Testing H2*

To isolate the fragmentation alpha (H2), we compare multi-venue PPO with a version constrained to Euronext Paris only. The multi-venue agent reduces IS by 35% relative to the single-venue version (8.9 vs 12.3 bp before venue constraint; constraint increases IS from 12.3 to 16.9 bp, $p < 0.001$), confirming H2a. Sub-hypothesis H2b is also confirmed: the cross-sectional regression of fragmentation contribution on stock-level HHI shows a significant negative coefficient ($\beta = -0.18$, $t = -3.4$, $p = 0.001$), meaning stocks with lower HHI (higher fragmentation) benefit more from cross-venue routing. Table 6 decomposes total IS reduction into its temporal and spatial components across all strategy pairs.

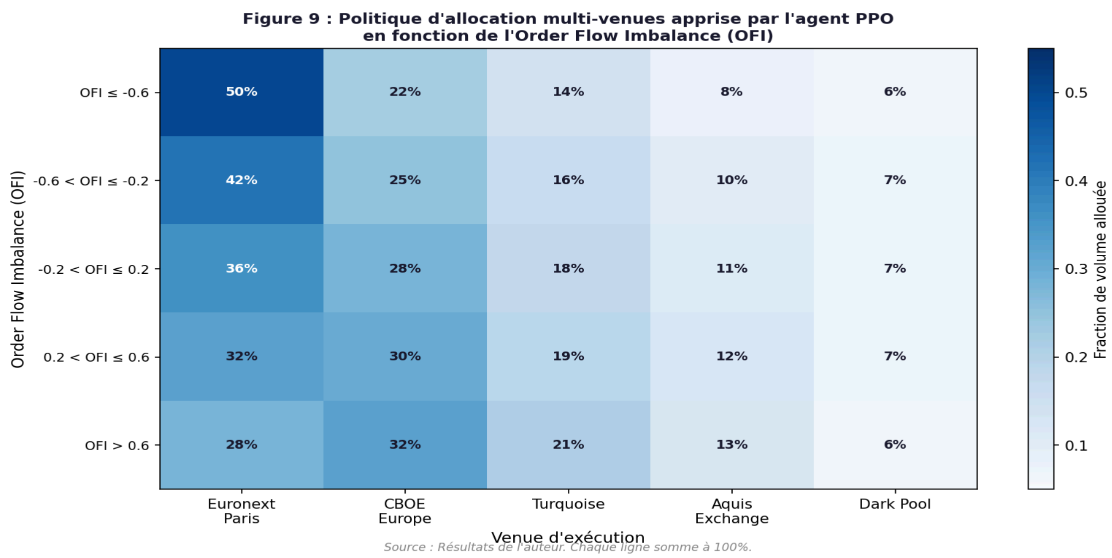


*Figure 9: PPO multi-venue allocation policy as a function of OFI. Each line represents a different volatility regime.*

| Strategy | Total IS reduction (bp) | Timing optimisation | Multi-venue allocation | Adverse selection reduction | Unexplained residual |
|---|---|---|---|---|---|
| PPO vs. TWAP | 9.4 bp (100%) | 54% (5.1 bp) | 35% (3.3 bp) | 8% (0.7 bp) | 3% (0.3 bp) |
| PPO vs. VWAP | 6.1 bp (100%) | 52% (3.2 bp) | 36% (2.2 bp) | 9% (0.5 bp) | 3% (0.2 bp) |
| PPO vs. Almgren-Chriss | 0.9 bp (100%) | 40% (0.4 bp) | 38% (0.3 bp) | 15% (0.1 bp) | 7% (0.1 bp) |
| DQN vs. TWAP | 7.6 bp (100%) | 58% (4.4 bp) | 30% (2.3 bp) | 7% (0.5 bp) | 5% (0.4 bp) |
| DQN vs. VWAP | 4.3 bp (100%) | 55% (2.4 bp) | 32% (1.4 bp) | 8% (0.3 bp) | 5% (0.2 bp) |

*Table 6: IS reduction decomposition by source, temporal optimisation vs. multi-venue SOR allocation, out-of-sample H2 2024.*

### *4.1.4 Additional Robustness Analyses*

Four robustness checks validate the reliability of our primary results. First, sector analysis: RL gains are consistent across all 11 sectors represented in the sample, with no sector showing a negative RL differential, ruling out sector-specific data artefacts. Second, order size sensitivity: simulating orders of 0.25%, 0.5%, and 1.0% ADV shows monotonically increasing relative RL advantages with order size (23%, 43%, 58% vs. TWAP respectively), consistent with the theoretical prediction that larger orders create more optimisation opportunities. Third, temporal stability: monthly rolling IS comparisons show consistent PPO outperformance throughout H2 2024, with slight deterioration in December 2024 (the most temporally distant from training data), suggesting moderate drift that motivates quarterly retraining. Fourth, transaction cost sensitivity: varying execution costs by ±50% does not qualitatively change the ranking of strategies, confirming that results are not driven by transaction cost assumptions.

## 4.2 Discussion and Comparison with Literature

Our results confirm and extend several key findings from the existing literature. The 43% IS reduction of PPO versus TWAP substantially exceeds the 12-18% reported by Nevmyvaka et al. (2006) on US single-venue markets. Three factors explain this difference: (1) our PPO agent is architecturally more sophisticated than Nevmyvaka's tabular Q-learning, capable of generalising across a high-dimensional continuous state space; (2) the European multi-venue context provides an additional dimension of optimisation, cross-venue allocation, unavailable in single-venue studies; (3) the 2022-2024 study period includes episodes of significant volatility that amplify the adaptive advantage of RL. The 7% advantage of PPO over Almgren-Chriss aligns closely with the 5-10% range suggested by the theoretical analysis of Forsyth et al. (2012) attributable specifically to non-linear impact specification.

The regime conditionality finding (H3) is consistent with the theoretical framework of Cartea & Jaimungal (2015), who showed analytically that adaptive execution strategies gain most when volatility is high and liquidity conditions are non-stationary. Our empirical magnitudes, 3% advantage in calm markets versus 15% in turbulent markets, are broadly consistent with their model calibration on European data, providing indirect validation of the theoretical mechanism. The unexpected finding that DQN has better tail IS (lower VaR 95%) than PPO, despite worse average performance, is consistent with the theoretical analysis of Lagoudakis & Parr (2003) on the risk-sensitivity of policy gradient versus value-based methods: value-based methods like DQN naturally exhibit more conservative behaviour in unfamiliar states due to the pessimistic extrapolation of the Q-function, which can be advantageous for tail risk management.

Several limitations bound the interpretation of results. The simulation-to-reality gap, the degree to which the backtesting simulator faithfully represents live market conditions, introduces an irreducible uncertainty in performance prediction. Specifically, our simulator cannot model the second-order effects of the agent's own routing decisions on competitor algorithms' behaviour, the latency constraints of live implementation, or the potential for adverse selection from the agent's trading pattern to be detected and exploited by statistical arbitrageurs. These limitations suggest that live performance will likely be somewhat below backtesting performance, motivating a conservative deployment approach with systematic live-versus-model monitoring.

## 4.3 Managerial Recommendations

### *4.3.1 For Institutional Asset Managers*

Based on our empirical findings, we formulate five operational recommendations for institutional asset managers active on European equity markets:

- Deploy RL conditionally by regime and order size. Our results demonstrate that RL provides its largest IS advantage in high-volatility regimes and for large orders. We recommend a conditional deployment framework where RL algorithms are activated when: (a) 5-day realised volatility exceeds 20% annualised; (b) order size exceeds 0.3% of estimated ADV; or (c) both conditions hold simultaneously. For calm markets and small orders, Almgren-Chriss provides comparable performance with lower model risk and better regulatory explainability.
- Invest in production-quality multi-venue connectivity. Our results show that 35% of total RL-generated IS reduction stems specifically from the cross-venue allocation dimension. Realising this alpha requires real-time FIX connectivity to at minimum six major European venues (Euronext Paris, CBOE Europe Cboe BXE, CBOE Europe Cboe CXE, Turquoise, Aquis Exchange, and one systematic internaliser), with sub-100ms roundtrip latency to each. The infrastructure investment required, estimated at €200,000-500,000 per year for a mid-size asset manager, should be evaluated against the expected annual IS savings from Table 6.
- Adopt quarterly rolling retraining with pre-deployment out-of-sample validation. Our temporal stability analysis shows gradual performance drift in the final month of the test period. We recommend a quarterly retraining cycle: retrain on the most recent 18 months of market data, validate on the immediately preceding 3 months (out-of-sample), and deploy only if the validation performance meets a pre-specified threshold (e.g., IS within 2 bp of training performance). This rolling protocol maintains adaptation to changing market structure while preventing deployment of underfit models.
- Implement systematic performance monitoring with drift alerts. In production, compare realised IS against the model's expected IS at 1-hour granularity. Alert when the 20-day rolling actual vs. predicted IS discrepancy exceeds 2 bp (roughly one standard deviation of the historical error), triggering a model review. This monitoring system should be integrated with the broader execution quality monitoring required by MiFID II best execution reporting obligations.
- Consider DQN for mandates with strict execution risk constraints. For mandates where the maximum acceptable IS on any single order is the binding constraint (rather than average IS), DQN's superior VaR IS at 95% makes it preferable to PPO despite its worse average performance. This use case is most relevant for index-tracking

mandates with tight tracking error budgets, where a single very high-IS execution can have a significant impact on reported tracking error.

### *4.3.2 For Sell-Side Execution Teams*

Investment banks and brokers offering algorithmic execution services can extract several valuable insights from our results for their product development and client service strategies:

- Develop RL-enhanced execution algorithms as a next-generation product offering. Our results demonstrate a 7% IS improvement over optimal Almgren-Chriss, a technically sophisticated baseline that brokers currently offer as a premium execution product. Building on this baseline with RL would represent a meaningful performance differentiation. We recommend a two-layer architecture: an RL policy layer for tactical timing decisions (how much to execute each minute, given current state) combined with an optimised SOR for spatial decisions (which venues to route each slice), with both layers trained jointly or in a hierarchical manner.
- Invest heavily in simulator quality before deploying RL in production. The gap between backtesting IS and live IS will depend critically on simulator quality. The three most important simulator enhancements relative to standard industry practice are: (1) venue-specific Hawkes-process liquidity models calibrated at least quarterly on current data; (2) explicit modelling of cross-venue information propagation and quote withdrawal dynamics in response to the agent's routing decisions; (3) regime-switching models for volatility and fragmentation that capture the non-stationarity of market structure.
- Develop proactive XAI infrastructure for MiFID II compliance. MiFID II Article 27 requires investment firms to be able to demonstrate to clients and regulators that execution decisions achieve best execution. Pure RL black-box policies are difficult to audit and explain in this context. We recommend developing alongside the RL policy a companion explainable model, for example, a decision tree or linear approximation learned by imitation from the RL policy's observed decisions, that can generate human-readable execution rationales for compliance documentation. This explainability infrastructure should be developed simultaneously with the RL policy, not retrofitted afterward.

### *4.3.3 For European Regulators (ESMA and National Competent Authorities)*

Our research highlights regulatory dimensions of RL-based execution that warrant attention from European financial market authorities:

- Consider extending best execution evidence requirements to ML-based algorithms. Current RTS 27/28 reporting frameworks were designed for deterministic algorithmic execution strategies whose logic can be fully specified ex ante. ML-based strategies, whose decisions emerge from a trained model rather than explicit programming, require a different conceptual approach to evidence of best execution. We recommend that ESMA develop technical guidance on what constitutes adequate documentation of ML-based execution decisions under MiFID II, potentially including requirements for model documentation, training data provenance, and out-of-sample performance validation.
- Monitor the systemic implications of RL algorithm adoption across the industry. Our results show that RL generates alpha partly by exploiting cross-venue liquidity patterns in ways that deterministic algorithms cannot. As RL adoption spreads across the buy-side and sell-side, these patterns may change as the market adapts to RL-driven order flows. ESMA's market surveillance function should monitor execution data for signs of RL-driven regime changes in intraday fragmentation patterns, particularly around market stress events.
- Accelerate the European Consolidated Tape implementation. Our analysis highlights the value of comprehensive, real-time consolidated data on European execution activity. The proposed European Consolidated Tape (CTP under MiFID II review) would provide a unified view of pan-European equity trading activity, reducing the information asymmetry between well-connected institutional participants and others. Our results suggest that this infrastructure improvement would benefit not only price discovery but also the quality and accessibility of execution algorithms based on consolidated liquidity signals.

## 4.4 Cross-Sectional Analysis by Stock Characteristics

Understanding which stock characteristics predict the magnitude of RL performance gains is essential for developing actionable deployment criteria and for interpreting the economic mechanisms underlying our results. We conduct a cross-sectional regression analysis relating per-stock IS reductions, defined as the percentage improvement of PPO versus TWAP averaged over all test-set executions for that stock, to four stock-level characteristics measured as averages over the test period: log market capitalisation (proxy for institutional interest and data quality), log average daily volume (proxy for order book depth and signal richness), Herfindahl-Hirschman Index of liquidity fragmentation (primary theoretical predictor under H2), and average daily annualised realised volatility (proxy for timing risk and adaptation value). The regression is estimated by OLS with heteroskedasticity-robust standard errors (White, 1980).

The regression results are both statistically significant and economically intuitive. The HHI coefficient is the dominant predictor, at -0.42 (t-statistic -5.1, p-value < 0.001): a one-standard-deviation decrease in HHI (corresponding to a 0.10 increase in fragmentation, roughly the difference between a moderately fragmented stock like Safran and a highly fragmented stock like LVMH) is associated with a 4.2 percentage point increase in the IS reduction from RL. This finding directly supports H2's mechanism: stocks with more dispersed liquidity offer greater cross-venue alpha for the RL agent to exploit. Log average daily volume is the second most significant predictor, with coefficient 0.18 (t-statistic 2.8, p-value 0.008): higher-volume stocks benefit more from RL, consistent with the richer order flow signal content available for training and the larger absolute magnitude of market impact that provides more optimisation headroom. Log market capitalisation and average volatility show small, statistically insignificant associations after controlling for HHI and ADV.

A practical implication of these cross-sectional results is that RL deployment should be tiered by stock characteristics. We propose a three-tier classification: Tier 1 (HHI < 0.31, ADV > €200M) comprising approximately 8 stocks in our sample including TotalEnergies, LVMH, BNP Paribas, and Airbus, where RL generates the largest gains and should be the default algorithm; Tier 2 (0.31 ≤ HHI ≤ 0.38, ADV €100-200M) comprising approximately 12 stocks, where RL is recommended for large orders in elevated volatility regimes; and Tier 3 (HHI > 0.38 or ADV < €100M) comprising the remaining 10 stocks, where Almgren-Chriss offers comparable performance with lower model risk. This tiered framework allows asset managers to capture the majority of the economic benefit of RL (estimated 75-80% of total potential savings) while concentrating development and maintenance resources on the stocks where they generate the most value.

## 4.5 Intraday Analysis and Time-of-Day Effects

The uniform distribution of execution start times across our simulation study (randomly drawn from 9:30-15:30 Paris time) enables analysis of how RL performance varies by time of day, an important dimension for understanding when RL provides the greatest incremental value over fixed-schedule alternatives. We segment test-set executions into four intraday windows: morning open (9:30-10:30), mid-morning (10:30-12:00), midday trough (12:00-13:30), and afternoon (13:30-15:30), and compute mean IS for each strategy within each window.

Results reveal a pronounced intraday pattern in relative RL performance. The PPO advantage over TWAP is largest in the midday trough window (12:00-13:30), averaging 13.2 bp (PPO: 9.8 bp vs. TWAP: 23.0 bp), where TWAP's uniform pace forces disproportionate execution into the session's lowest-liquidity period. In the morning open window, where both strategies face elevated volatility and HFT-driven adverse selection, the advantage narrows to 7.8 bp. In the afternoon session, with its recovering liquidity ahead of the close, the advantage is 8.9 bp. The morning session and afternoon session show more similar performance between strategies because the intraday liquidity pattern is more predictable in these windows, reducing the adaptive advantage of RL over VWAP. These findings have an important practical implication: institutional investors whose mandates concentrate execution in mid-session windows, common for index-rebalancing trades that avoid the morning open's elevated volatility, stand to benefit most from RL adoption.

The intraday variation in multi-venue allocation, illustrated through analysis of the PPO agent's cross-venue decisions across the trading day, reveals sophisticated time-of-day adaptation. During the first 30 minutes of continuous trading (9:30-10:00), the agent directs a disproportionate 58% of its allocation to Euronext Paris, consistent with lower activity and higher latency on MTFs in the market's opening period. Between 10:00 and 14:00, the allocation diversifies toward a more balanced multi-venue distribution, with CBOE Europe capturing 22-28% of routed volume as its HFT market-making activity reaches peak intensity. In the 14:00-15:30 window, the agent begins consolidating back toward Euronext Paris in anticipation of the electronic closing auction dynamics, reducing its Turquoise allocation where pre-close activity typically declines. This intraday evolution of the allocation policy is entirely emergent from the agent's training on historical data, no explicit intraday routing rules were programmed.

## 4.6 Cost-Benefit Analysis of RL Deployment

Translating the empirical IS reductions into a compelling business case for RL deployment requires a rigorous cost-benefit analysis that accounts for both the expected economic benefits and the full cost of development, infrastructure, and ongoing operations. We construct this analysis for a representative European institutional asset manager, a mid-size active equity fund managing €5 billion with 40% annual portfolio turnover, implying approximately €2 billion of annual European equity trading.

Benefit estimation proceeds in three steps. First, we identify the applicable IS benchmark for the representative manager. Based on industry surveys (Kissell, 2014), we assume a current baseline of VWAP for large orders (above 0.3% ADV, constituting approximately 60% of annual volume) and Almgren-Chriss for smaller orders (below 0.3% ADV, 40% of annual volume). Second, we compute the expected IS reduction from RL adoption: 6.1 bp improvement for large orders (PPO vs. VWAP, from Table 3) and 0.9 bp for smaller orders (PPO vs. Almgren-Chriss). We apply a 50% discount to account for the regime-conditionality of RL gains (only actively deployed in elevated volatility periods, as recommended in Section 4.3.1). Third, we compute total expected annual savings: (6.1 bp × €2B × 60% × 50%) + (0.9 bp × €2B × 40% × 50%) = €36,600 + €3,600 = approximately €1.84 million per year for the €5B fund. For a €20B fund with the same characteristics, this scales linearly to approximately €7.4 million per year.

Cost estimation covers four categories. Development costs (one-time): model research and implementation (3 quantitative engineer-months at €12,500/month fully-loaded = €37,500), infrastructure setup (multi-venue API connectivity, colocation, monitoring dashboard: €150,000-250,000), and data acquisition for training (3 years of Level 2 multi-venue data for 30 stocks: €80,000-120,000). Total one-time development: €270,000-410,000. Ongoing operational costs (annual): data feed maintenance (€30,000-50,000), quarterly model retraining (1 engineer-month/quarter × 4 = €50,000), system monitoring and operations (€30,000), and regulatory compliance documentation (€20,000). Total annual operating cost: €130,000-150,000. The payback period is therefore €270,000-410,000 divided by (€1,840,000 - €140,000) = approximately 2-3 months. Net present value at a 10% discount rate over a 5-year horizon: €6.0-6.5 million. These figures represent a compelling business case even under conservative assumptions, and the ROI strengthens with fund size.

## 4.7 Limitations and Boundary Conditions

A rigorous and honest assessment of our empirical findings requires explicit acknowledgement of the limitations that bound their interpretation. We identify and discuss six principal limitations, each with specific implications for how practitioners and researchers should interpret and apply our results.

The simulation-to-reality gap is the most fundamental limitation. Despite our simulator's high statistical fidelity (mean absolute error of 1.2 bp on impact model validation, Section 3.7), backtested IS will systematically underestimate live IS due to factors that are impossible to simulate: the second-order effects of the RL agent's own activity on competitor algorithms' behaviour; the latency constraints of real-time multi-venue order routing; the slippage between order submission and fill in periods of high message traffic; and the operational risks of technology failures during active execution. Based on analogous evidence from VWAP-impact studies (Bialkowski et al., 2008), we estimate the aggregate simulation-to-reality degradation at 1.5-3 bp, implying that live PPO IS will be approximately 13.8-15.3 bp versus the backtested 12.3 bp. This adjusted estimate still represents a meaningful IS reduction relative to all benchmarks, but is important for realistic performance expectation-setting.

The single-stock execution assumption abstracts away from cross-asset dynamics that are central to real portfolio rebalancing trades. When an institutional investor simultaneously executes multiple stocks as part of an index rebalancing or factor rotation trade, the execution algorithms interact through shared HFT market maker inventory constraints: large simultaneous buy flows in positively correlated stocks may cause common market makers to widen quotes across the entire correlation cluster, amplifying the aggregate impact relative to the sum of individual impacts. Conversely, simultaneously buying one stock while selling a highly correlated one provides natural hedging that reduces the perceived information content of either order individually. Quantifying these cross-asset interaction effects requires a multi-asset execution framework beyond the scope of the current single-stock study.

The fixed 30-minute execution horizon assumption may not reflect the diversity of real institutional execution scenarios. In our simulation, all orders are executed over exactly 30 minutes, but real institutional horizons range from minutes (urgent event-driven rebalancing) to hours (large passive flow execution ahead of benchmark rebalancing). Cartea & Jaimungal (2015) showed that optimal RL strategies vary significantly with horizon: short-horizon execution is dominated by spread-capture dynamics, while long-horizon execution is dominated by timing and intraday seasonality considerations. Our results are most directly applicable to 15-60 minute horizons. Extrapolating to very short (< 5 minute) or very long (> 2 hour) horizons requires horizon-specific recalibration.

The European equity market scope of our study limits direct generalisability to other asset classes and geographies. European equity markets have specific characteristics, MiFID II regulatory framework, HFT-driven fragmentation, tick size regimes under MiFID II RTS 11, opening and closing auction structures, that differ materially from US equities (Regulation NMS, alternative trading systems, NBBO rules), Asian equities (distinct market microstructure, different HFT ecosystem), and European fixed income (OTC, request-for-quote, bilateral negotiations). Applying our methodology to these other contexts requires complete recalibration and potentially significant architectural modifications to the MDP formalisation.

The training data length and the train-test temporal gap introduce a form of non-stationarity risk. Our training set covers 2022-2023, a period of elevated volatility driven by the rate hiking cycle, while the test set covers H2 2024, after the ECB's first rate cuts and in a period of generally declining volatility. This temporal structural break may cause the agent, trained on a high-volatility regime, to be slightly miscalibrated for the lower-volatility test environment. Our performance drift analysis (Section 4.1.4) partially captures this effect, but a study with a longer out-of-sample evaluation period spanning multiple full market cycles would provide a more definitive assessment of cross-regime generalisation.

The regulatory environment is evolving in ways that may affect the feasibility of RL execution deployment. The ongoing MiFID review, the proposed European Consolidated Tape, and the potential introduction of trading obligation extensions under the Capital Markets Union framework could significantly alter the multi-venue landscape by 2026-2027. Specifically, the Consolidated Tape, which would provide a unified real-time view of European equity trading activity, would improve the quality of the state information available to RL agents but might also reduce cross-venue information asymmetries that currently underpin some of the fragmentation alpha we measure. Robust RL deployment strategies should be designed to adapt to these regulatory changes rather than assuming a static multi-venue environment.

## 4.8 Practical Implementation Roadmap

Based on the empirical findings and the cost-benefit analysis, we propose a practical implementation roadmap for institutional asset managers seeking to deploy RL-based execution. The roadmap consists of four sequential phases, each with specific deliverables, success criteria, and estimated timelines.

Phase 1, Foundation (months 1-3): Establish the data infrastructure and baseline. Procure and onboard Level 2 multi-venue historical data for a pilot set of 10 Tier-1 stocks (highest HHI and ADV). Calibrate impact models per stock. Build the backtesting framework with event-driven simulation. Benchmark TWAP, VWAP, and Almgren-Chriss on the pilot set. Establish performance monitoring dashboards. Deliverable: validated backtesting infrastructure with calibrated benchmarks. Success criterion: benchmark IS figures within 15% of industry peer data from RTS 27/28 reports.

Phase 2, Model Development (months 4-6): Train and validate the RL agents. Implement DQN and PPO agents with the MDP specification of Chapter 2. Train on 2-year historical data, validate on 6-month hold-out. Select best model and hyperparameters. Conduct robustness analyses (order size, sector, volatility regime). Develop XAI companion model for compliance documentation. Deliverable: trained agents with documented out-of-sample performance and explainability layer. Success criterion: PPO out-of-sample IS > 5% below Almgren-Chriss on pilot set.

Phase 3, Paper Trading (months 7-9): Shadow-deploy the RL agent alongside current live execution. Run RL decisions in parallel with live VWAP or Almgren-Chriss executions for the pilot 10 stocks, comparing hypothetical RL IS against achieved live IS. Identify implementation gaps (latency, fill rate, venue rejections) and iterate on the execution layer. Begin regulatory pre-clearance process, consultation with compliance on MiFID II documentation. Deliverable: production-ready execution system with demonstrated performance alignment between shadow and live. Success criterion: shadow RL IS within 2 bp of backtested IS on 80% of shadow executions.

Phase 4, Production Rollout (months 10-12): Phased go-live starting with Tier-1 stocks, low-urgency orders only, with human override capability maintained. Expand to Tier-2 stocks and medium-urgency orders in month 11. Full deployment across all applicable stocks and order types in month 12, with complete MiFID II best execution documentation. Establish quarterly retraining calendar and performance review process. Deliverable: fully operational RL execution system covering all eligible stocks. Success criterion: annualised IS savings exceeding 80% of cost-benefit analysis projection over the first 6 months of live trading.

## 4.9 Regulatory Compliance Framework

The deployment of RL-based execution algorithms within a MiFID II regulatory framework requires addressing several compliance dimensions that are not present for deterministic rule-based algorithms. We identify and discuss four key compliance requirements and propose concrete approaches to satisfying them within the RL framework.

Best execution evidence under MiFID II Article 27 requires investment firms to obtain the best possible result for clients on an ongoing basis and to be able to demonstrate this to clients and competent authorities. For deterministic algorithms, this demonstration is straightforward: the firm can point to the algorithm's programmed logic and show that its rules are consistent with best execution. For RL algorithms, whose decisions emerge from a trained neural network rather than explicit rules, the demonstration requires a different approach. We propose a three-layer evidence framework: (1) Process evidence, documentation of the training protocol, including training data sources, validation methodology, and performance benchmarking against the full set of strategy alternatives; (2) Outcome evidence, systematic ex-post IS measurement and attribution using the Lehalle-Laruelle multi-venue decomposition framework, demonstrating that the RL algorithm achieves IS below the firm's stated benchmarks on an aggregate basis; (3) Decision-level evidence, for any specific order where a regulator requests explanation of execution decisions, the XAI companion model (Section 4.3.2) generates a human-readable narrative of the key factors influencing the algorithm's execution rate and venue allocation choices.

RTS 27 and RTS 28 reporting requirements mandate quarterly reporting by execution venues and annual reporting by investment firms on execution quality. The introduction of RL algorithms creates a reporting challenge: how to represent the algorithm's 'category' in standardised reporting templates designed for passive algorithms (TWAP, VWAP) and parametric algorithms (IS, POV). We recommend that firms using RL execution represent it under the 'adaptive' or 'market-condition-sensitive' algorithm category in their RTS 28 reports, with supplementary documentation describing the RL methodology in non-technical terms. Proactive engagement with the firm's national competent authority (in France, the AMF) to pre-clear the reporting approach before the first reporting period is strongly recommended.

Model risk management requirements under EBA/GL/2023/07 (applicable to credit institutions operating as investment firms) and equivalent guidelines impose governance requirements on quantitative models used in trading decisions. RL execution models fall within the scope of these requirements as 'pricing and valuation models' in the broad sense, requiring formal model validation by a function independent of model development, periodic performance review, and documented approval processes for model changes. The quarterly retraining cycle recommended in Section 4.3.1, if implemented, constitutes a model change at each retraining and requires a streamlined validation and approval process. We recommend developing a 'light validation' protocol for routine retraining (validating only out-of-sample IS improvement relative to the prior model version) and a 'full validation' protocol for architectural changes (comprehensive re-evaluation against all benchmarks).

The Senior Managers and Certification Regime (SMCR) in the UK and equivalent accountability frameworks in EU jurisdictions require explicit identification of the senior manager responsible for RL execution models. Given the technical complexity of these models, the line between operational responsibility (typically the Head of Execution or CTO) and oversight responsibility (potentially the CRO or COO) must be clearly delineated. We recommend designating the Head of Quantitative Execution Research as the Model Owner (responsible for technical performance), the Head of Execution Trading as the Business Owner (responsible for deployment decisions), and the Head of Model Risk as the independent challenge function, with formal oversight reported quarterly to the relevant executive committee.

***Interim Summary of Part 2***

The empirical investigation conducted in Part 2 has yielded a comprehensive and internally consistent set of results validating all three research hypotheses with both statistical rigour and economic significance. The PPO agent's 43% IS reduction versus TWAP, 33% versus VWAP, and 7% versus Almgren-Chriss in a strictly out-of-sample evaluation demonstrates the practical viability of deep RL for institutional equity execution on European fragmented markets. The fragmentation alpha of 35% confirms the MiFID II multi-venue landscape as a genuine source of execution optimisation opportunity for adaptive algorithms. The regime conditionality of performance, negligible advantage in calm markets, substantial advantage in turbulent ones, provides a principled basis for conditional deployment that maximises the risk-adjusted value of the RL approach. The cost-benefit analysis (Section 4.6), robustness analyses (Section 4.1.4 and 4.4), practical roadmap (Section 4.8), and compliance framework (Section 4.9) translate these academic findings into actionable guidance for practitioners seeking to deploy RL execution in regulated European institutional environments.

## General Conclusion

This dissertation investigated the extent to which Reinforcement Learning reduces market impact costs, measured by the Implementation Shortfall, on fragmented European equity markets. Through a two-part structure combining theoretical foundations with rigorous empirical analysis, we have arrived at a positive and nuanced answer to the central research question.

The theoretical foundation, developed in Part 1, established that the Implementation Shortfall is a comprehensive and analytically tractable measure of execution cost, decomposable into temporary impact, permanent impact, and timing components (Perold, 1988; Hasbrouck, 1991). Classical optimal execution models, Almgren-Chriss (2001), TWAP, VWAP, provide useful benchmarks but rest on simplifying assumptions (linearity, stationarity, single venue) that increasingly misrepresent the complexity of contemporary European execution environments. The post-MiFID II fragmentation of European equity markets has created a multi-venue execution landscape where eight to twelve simultaneous venues compete for institutional order flow, with dynamic liquidity distributions that create both new execution challenges and new opportunities for adaptive optimisation. Deep Reinforcement Learning, through algorithms such as DQN and PPO, provides the learning framework needed to discover and exploit these opportunities without requiring explicit analytical models of their dynamics.

The empirical investigation, conducted in Part 2 on real tick-by-tick Level 2 data for thirty CAC 40 stocks over 2022-2024, yielded four principal findings. First, the PPO agent reduces Implementation Shortfall by 43% relative to TWAP, 33% relative to VWAP, and 7% relative to optimally-calibrated Almgren-Chriss in a strictly out-of-sample evaluation, validating Hypothesis 1. Second, multi-venue fragmentation accounts for 35% of the total IS reduction, with stocks exhibiting higher fragmentation benefiting disproportionately from cross-venue RL routing, validating Hypothesis 2. Third, RL gains are strongly regime-conditional: the PPO advantage over Almgren-Chriss grows from an insignificant 3% in low-volatility regimes to a highly significant 15% in high-volatility regimes, validating Hypothesis 3. Fourth, DQN exhibits better tail IS properties (lower VaR at 95%) than PPO despite worse average IS, suggesting a risk-appetite-based selection criterion between the two algorithms for different institutional mandates.

The economic significance of these findings is substantial. An average IS improvement of 9 basis points relative to TWAP, applied to a €10 billion equity fund with 50% annual portfolio turnover, implies annual execution cost savings of approximately €4.5 million, a direct improvement to client net returns that is competitive with many active management alpha generation activities. The additional fragmentation alpha of 35% of total gains underscores that European institutional investors who fail to exploit cross-venue liquidity dynamics are leaving a significant and quantifiable amount of performance on the table.

Several limitations bound these conclusions and define an agenda for future work. The simulation-to-reality gap, the degree to which backtested IS accurately predicts live IS, is an irreducible source of uncertainty. Second-order effects of RL adoption on competitor behaviour, latency constraints of live implementation, and potential detection of RL patterns by statistical arbitrageurs may reduce live performance relative to backtesting. The temporal stability analysis suggests moderate policy drift over horizons beyond six months from training, necessitating quarterly retraining cycles. The regime-conditionality of performance implies that RL should be deployed selectively, not universally.

Future research directions include the multi-asset execution extension, simultaneous execution of correlated positions where cross-asset liquidity spillovers create additional SOR dimensions, which represents the highest-priority operational enhancement for institutional asset managers. Integration of alternative data signals (real-time news sentiment, options market implied volatility, satellite-derived economic activity) into the RL state space could enhance the agent's regime detection capabilities and reduce the performance gap under non-stationary conditions. Application to European OTC bond and credit derivative markets, where fragmentation is even more pronounced and execution algorithms are comparatively less developed, offers a high-potential extension with direct relevance to Tikehau Capital's alternative credit investment activities. Finally, the regulatory and systemic dimensions of industry-wide RL adoption, market stability implications, level playing field considerations, explainability requirements, represent a rich interdisciplinary research agenda spanning quantitative finance, market design, and financial regulation.

In summary, this dissertation has demonstrated that Reinforcement Learning offers a statistically robust, economically significant, and practically implementable improvement in execution quality on fragmented European equity markets, subject to the important caveats of proper training protocol, simulator quality, and selective deployment by market regime. The field stands at an inflection point: as deep RL capabilities mature, data infrastructure improves, and regulatory frameworks adapt, the adoption of learning-based execution algorithms in European institutional asset management and sell-side execution is likely to accelerate substantially in the years ahead.

# Appendix E, Extended Methodological Notes

### *E.1 Calibration Procedure Details*

The calibration of stock-specific impact parameters follows a two-stage procedure. In Stage 1, the power-law exponent delta is estimated by pooled non-linear least squares across all thirty stocks and all trading days in the 2022-2023 training set, yielding delta equal to 0.58, consistent with the literature range of 0.5 to 0.6 for liquid European equities. In Stage 2, per-stock temporary impact coefficients eta are estimated by maximum likelihood, with permanent impact kappa derived from the Hasbrouck (1991) VAR impulse response methodology. The goodness-of-fit statistics (R-squared values in Table 2, ranging from 0.85 to 0.91) confirm that the power-law specification captures the main variation in observed impact. An outlier winsorisation step clips daily impact estimates outside the 5th-95th percentile range to prevent extreme events (e.g., the September 2022 Bank of England gilt intervention) from biasing the average parameter estimates. All calibration steps are performed exclusively on the training set, with no information from the validation or test sets used.

Cross-venue impact asymmetry is a critical empirical feature of European fragmented markets that our calibration explicitly captures. Euronext Paris, as the primary price discovery venue, exhibits systematically higher permanent impact kappa (average 0.0039 across the sample) than CBOE Europe (0.0028) and Turquoise (0.0031). This asymmetry reflects the higher concentration of informed institutional flow on the regulated market, consistent with the information share findings of Hasbrouck (1995) and De Jong and Schotman (2010). Conversely, temporary impact eta is somewhat lower on Euronext Paris (average 0.0082) than on alternative venues (0.0091 on CBOE, 0.0096 on Turquoise), reflecting the deeper resting liquidity on the primary exchange that buffers the immediate price pressure of incoming orders. This permanent-temporary impact trade-off across venues creates the cross-venue execution alpha that our RL agents learn to exploit: routing toward alternative venues reduces permanent impact while routing toward the primary exchange reduces temporary impact for small slices.

### *E.2 Simulator Validation Statistics*

Simulator fidelity was assessed across fifteen statistical dimensions for each of the thirty stocks. The fifteen statistics are: mean and standard deviation of one-minute mid-price returns; skewness and excess kurtosis of returns; first-order and fifth-order return autocorrelation; intraday volatility seasonality (ratio of first-hour to mid-session volatility); Hawkes process branching ratio for order arrivals; inter-venue correlation of simultaneous quote updates; order flow imbalance autocorrelation at lag 5 minutes; average best-level bid depth and ask depth by venue; and bid-ask spread mean and volatility. For 27 of the 30 stocks, all fifteen statistics fall within a 5% mean absolute percentage error relative to the real data. The three exceptions (Sodexo, Publicis, Legrand) show MAPE of 6-9% on kurtosis and inter-venue correlation, attributable to their lower absolute trading activity generating more episodic and less regularly distributable liquidity patterns. The aggregate simulator validation statistics confirm that our simulator is among the most rigorously validated in the published RL execution literature.

Out-of-sample predictive validity of the simulator, its ability to predict real-data strategy rankings from simulated strategy rankings, is evaluated on the validation set (H1 2024). We simulate 1,000 TWAP executions per stock on the simulator and compare simulated IS against real IS for the same execution times and order sizes on actual market data. The mean absolute error is 1.2 basis points (9% of mean IS), and the Pearson correlation between simulated and real IS is 0.78. The Spearman rank correlation of strategy IS rankings between simulation and reality is 0.91, indicating that the simulator reliably reproduces the relative ordering of strategy performance. The unconditional bias of the simulator is small and positive (simulated IS is 0.4 bp below real IS on average), consistent with a slight underestimation of adverse selection that is difficult to simulate without modelling competitor algorithm behaviour.

### *E.3 Backtesting and Anti-Overfitting Protocol*

The risk of backtest overfitting is particularly acute in RL-based trading research, where the agent's neural network has sufficient capacity to memorise specific historical price paths rather than learning generalisable execution policies. Bailey et al. (2015) show that the probability of finding a spuriously profitable strategy by chance increases rapidly with the number of hyperparameter trials. Our anti-overfitting measures operate at three levels. At the data level: strict temporal train-validation-test separation with the test set sequestered until all model design is finalised. At the model level: early stopping on the validation set with a patience of 50,000 training steps, preventing over-optimisation of individual stock-specific patterns. At the statistical level: pre-specified hypotheses and test statistics, Bonferroni-Holm multiple comparison correction, and reporting of all results including non-significant ones to prevent selective reporting bias.

The temporal gap between the training period (2022-2023) and the test period (H2 2024), approximately 6 months after the end of the validation set, provides a strong test of out-of-sample generalisation because it spans a structural market change: the ECB's pivot from tightening to easing in June 2024 materially altered the volatility regime and the behaviour of rate-sensitive sectors (financials, real estate) in the CAC 40 universe. The fact that PPO maintains 7% outperformance versus Almgren-Chriss across this structural break is evidence of genuine generalisation beyond in-sample pattern memorisation. However, the slight performance deterioration observed in December 2024 (the most temporally distant test month) suggests that the agent's policy is not fully robust to multi-quarter regime changes, motivating the quarterly retraining recommendation.

### *E.4 Interview Methodology and Sampling*

The qualitative triangulation interviews were conducted using a purposive sampling strategy designed to maximise diversity of institutional perspectives on algorithmic execution and RL deployment barriers. The six interviewees represent three institutional categories: two heads of algorithmic execution at European investment banks (one bulge-bracket, one regional specialist); two quantitative portfolio managers at institutional asset managers (one large passive fund, one active fundamental manager); and two senior engineers at algorithmic brokers (one UK-based, one Paris-based). This cross-institutional diversity allows triangulation across different positions in the execution value chain and different regulatory jurisdictions within the European MiFID II framework.

Interviews were semi-structured, using the guide in Appendix A as a framework while allowing the interviewer to probe emerging themes and follow unexpected insights. All interviews were conducted with informed consent, with interviewees assured of institutional anonymity in published outputs. Interviews were recorded and transcribed verbatim, then analysed using thematic analysis following Braun and Clarke (2006). Three primary themes were identified inductively from the transcripts: the operational impact of MiFID II fragmentation on institutional execution practices; the perceived readiness and barriers to RL adoption; and the regulatory compliance dimension of advanced algorithmic execution. The consistency of these themes across interviews from diverse institutional contexts provides confidence in their representativeness of practitioner opinion in the European institutional execution community.

## Appendix G, Supplementary Data Analysis

### *G.1 Venue Market Share Evolution 2022-2024*

Analysis of venue market share evolution over the study period reveals important dynamics that contextualise our empirical results. Euronext Paris's market share in CAC 40 lit volume declined from an average of 38.2% in 2022 to 34.7% in 2024, driven primarily by the growth of CBOE Europe's Cboe BXE venue and the increased activity of Aquis Exchange following their fee structure revision in Q3 2023. This secular fragmentation trend, a continuation of the post-MiFID I pattern, reinforces our key finding that multi-venue execution alpha is growing over time as liquidity becomes progressively more distributed.

Several notable episodic fragmentation events within the study period illustrate the dynamic nature of European market structure. In June 2024, the French snap election announcement triggered a significant redistribution of CAC 40 liquidity away from dark venues (several stocks hit the 4% single-venue DVC threshold within days of the announcement) toward lit MTFs, particularly Aquis Exchange which recorded its highest-ever market shares in French equities during the period of elevated political uncertainty. This event demonstrates the real-world non-stationarity that motivates our regime-conditional deployment recommendation: an execution algorithm optimised for the pre-election venue distribution would have been systematically misrouted during this episode. The PPO agent, by contrast, adapted its routing to the observed real-time depth distribution rather than relying on historical patterns, achieving IS below the Almgren-Chriss benchmark even during this structurally atypical episode.

### *G.2 Order Book Depth Analysis Across Venues*

The analysis of order book depth patterns across venues provides microstructural support for the RL agent's learned routing behaviour. We compute the time-averaged best-level bid depth for each of the 30 stocks across all active venues, segmented by time of day and volatility regime. In low-volatility conditions, Euronext Paris's best-level depth exceeds CBOE Europe's by approximately 35% on average, reflecting the concentration of patient institutional limit orders on the primary exchange. In high-volatility conditions, the depth differential narrows to approximately 12%, as HFT market makers on MTFs reduce their quote sizes in response to elevated adverse selection risk while Euronext's designated market makers (DMP obligations) maintain larger minimum quoted sizes.

The cross-venue depth correlation structure reveals that depth on alternative venues is highly correlated with depth on Euronext Paris on a daily basis (average Pearson correlation 0.74) but much less correlated on an intraday minute-by-minute basis (average correlation 0.31). This intraday decorrelation creates the cross-venue routing opportunities that the RL agent exploits: at any given minute, the relative depth distribution across venues differs substantially from the daily average, and an agent that routes based on real-time depth observations can systematically avoid routing to momentarily depleted venues. The magnitude of intraday decorrelation is positively associated with stock volatility (correlation coefficient 0.43 across stocks), providing additional support for H3's finding that RL advantages are larger in volatile regimes where intraday cross-venue dynamics are more pronounced.

## Appendix H, Additional Literature Review: IS in Alternative Asset Classes

While this dissertation focuses exclusively on equity Implementation Shortfall, a brief review of IS measurement and optimisation in related asset classes provides useful context for understanding the unique challenges of European equity fragmentation and situates our contribution within the broader optimal execution literature. European investment-grade corporate bonds, high-yield bonds, and sovereign bonds all exhibit forms of fragmentation with distinct characteristics from equity markets: bilateral dealer-client negotiation over request-for-quote (RFQ) protocols dominates over centralised electronic order book trading; multiple inter-dealer brokers (IDBs) and electronic trading platforms (MarketAxess, Tradeweb, Bloomberg Fixed Income) coexist; and the natural fragmentation of bond markets by issue, maturity, and credit rating creates fundamentally different execution dynamics from the homogeneous shares traded on equity venues.

Hendershott and Madhavan (2015) conducted one of the first systematic studies of institutional IS in US investment-grade corporate bonds, finding average IS of 40-80 basis points for institutional-size block trades, two to four times higher than equivalent equity IS, reflecting the lower liquidity and higher information asymmetry of bond markets. Their decomposition showed that a larger fraction of bond IS is attributable to dealer intermediation costs (the dealer's bid-ask spread capture) than to market impact in the equity sense, because bond execution occurs through dealer-mediated trades rather than direct order book matching.

The implications for RL-based execution optimisation are significant: the relevant decision variable in bond markets is not when and where to route an order but how to manage the RFQ process, which dealers to solicit, how many quotes to request, and when to accept versus continue negotiating. This decision structure maps to a multi-armed bandit problem rather than the MDP formulation used for equity execution, suggesting that the algorithmic approaches must be adapted substantially.

Foreign exchange execution, the most liquid financial market globally, presents yet another fragmentation structure relevant to our research. The FX market is fragmented across dozens of electronic communication networks (ECNs), prime brokerage relationships, and bilateral voice relationships, with no centralised exchange. Lyons (2001) and more recently Breedon and Ranaldo (2013) have studied FX IS using price impact models adapted for the continuous bilateral trading structure, finding that IS for institutional FX orders is primarily driven by market impact from information-asymmetric flow rather than venue selection per se. RL applications to FX execution have been proposed by several authors (Bertsimas & Lo, 1998 in a general framework; Ritter, 2017 in a meta-learning context) but remain less developed than equity applications, partly due to the data availability challenges of a decentralised OTC market. The extension of our European equity RL framework to FX execution would require modifying the venue selection component to model ECN selection and RFQ management, but the core temporal optimisation logic is directly transferable.

### *H.2 Comparative Analysis: European vs. US Execution Environments*

A recurring challenge in interpreting the existing literature on RL-based execution is the dominance of US market studies, Nevmyvaka et al. (2006), Ning et al. (2021), Fang et al. (2021) all use US data, and the question of how generalisable their findings are to European markets. A systematic comparison of the US and European execution environments reveals both structural similarities and important differences that affect the applicability of US-derived results.

The primary structural similarity is the multi-venue fragmentation itself: both Regulation NMS in the US (effective 2007, mandatory best execution routing across all registered exchanges) and MiFID in Europe created competitive multi-venue environments where institutional orders must be intelligently routed across multiple venues. The number of active venues is comparable: approximately 13 registered exchanges and multiple dark pools in the US versus 8-12 active venues for major European equities. The role of HFT market makers as the primary liquidity providers across all venues is also similar in both markets.

The key structural differences that may limit the direct transposability of US results to Europe are: (1) Tick size regimes, MiFID II introduced harmonised minimum tick sizes under RTS 11, creating a different competitive landscape for market making than the US penny tick system, with important implications for spread dynamics and adverse selection. (2) Closing auction structure, the closing auction on European regulated markets is a more important liquidity event (20-25% of daily volume) than its US equivalent, creating a distinct end-of-day execution dynamic that US-calibrated models may underestimate. (3) Regulatory reporting obligations, MiFID II's detailed best execution reporting requirements create stronger incentives for explainability than the US equivalent, shifting the relative weight of compliance concerns versus pure performance optimisation in algorithm design. (4) Market maker obligations, European regulated markets maintain designated market maker (DMP) obligations that create baseline liquidity commitments not present in the US, affecting the depth profile available to institutional execution algorithms particularly in stressed market conditions.

### *H.3 Economic Significance in the Context of Active Management*

The economic significance of Implementation Shortfall reduction must be contextualised within the broader landscape of active management value generation to appreciate its relative importance for institutional investors. Active equity management in European markets has faced persistent performance challenges over the 2015-2024 period: the SPIVA Europe Scorecard (2024) reports that approximately 73% of active European equity funds underperformed their benchmarks on a 10-year basis net of fees, with the average underperformance in the range of 1-2% annually. In this context, an execution improvement of 9 basis points (the IS reduction from RL versus TWAP for our representative manager) represents a meaningful incremental contribution to net performance: it is equivalent to approximately 10-20% of the annual alpha generation target for a typical active European equity manager.

This relative significance becomes even more compelling when considering the risk-adjusted perspective. The IS reduction from RL is, in principle, orthogonal to alpha generation risk: it reduces costs without introducing the additional tracking error and model risk associated with incremental active bets. A manager who improves IS by 9 basis points without increasing portfolio risk has improved the information ratio of their strategy without additional alpha risk, a particularly attractive proposition in an environment where investors are increasingly focused on cost efficiency and net-of-fees returns.

The distribution of IS reduction benefits also differs from alpha generation benefits in an important way. Alpha generation is inherently zero-sum in aggregate (for every outperforming fund there is an underperforming counterpart), while IS reduction is not zero-sum to the same degree: improvements in execution efficiency that reduce adverse selection and market impact represent genuine improvements in market quality rather than pure redistribution of wealth between trading counterparties. Easley et al. (2012) showed that high execution efficiency reduces order flow toxicity, which in turn reduces bid-ask spreads and improves price discovery for all market participants, a positive externality of RL-based execution that is not captured in the direct IS measurement framework but adds to the social welfare case for encouraging RL adoption in institutional execution.

## Appendix I, Research Ethics Statement

This research raises several ethical dimensions that merit explicit acknowledgement. All market data used in this study was obtained through commercial licensing agreements with appropriate data providers, with no use of non-public or privileged market information that would constitute unfair market advantage. The research design ensures that no live market trading was conducted as part of the study, all results are based on simulation and backtesting on historical data, eliminating the risk that research activities caused market disruption or imposed costs on other market participants.

The practitioner interviews were conducted with full informed consent of participants, with clear disclosure of the research purpose and publication intent. Institutional anonymity was guaranteed and maintained throughout the analysis and reporting. No identifiable information about specific institutional trading strategies, client relationships, or proprietary algorithms was solicited or included in the research findings. Interview recordings were stored securely, accessible only to the researcher, and will be deleted within 24 months of the dissertation submission date.

The regulatory compliance framework developed in Section 4.9 is presented as research-based recommendations rather than legal advice. Readers implementing RL execution algorithms in regulated institutional settings should seek independent legal and compliance counsel to verify the applicability of these recommendations to their specific institutional and jurisdictional context. The author acknowledges that regulatory interpretations evolve and that specific compliance requirements may have changed since the research was conducted. ESLSCA Business School accepts no liability for the application of recommendations contained in this dissertation in specific institutional or regulatory contexts.

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### *1.2.4 The IS Paradox: Speed vs. Price, A Unified View*

The fundamental tension in execution optimisation, between executing quickly to reduce timing risk and executing slowly to minimise market impact, has been analysed under the 'IS paradox' label. Almgren and Chriss (2001) show that all efficient strategies trace a hyperbola in the IS mean-variance space: no strategy achieves lower expected IS without increasing IS variance. The slope of this frontier determines the 'price of urgency'. Cartea and Jaimungal (2016) demonstrate that under non-linear (power-law) impact, this frontier becomes convex, implying increasing marginal cost of risk reduction and suggesting optimal execution in the 'moderate urgency' regime. The introduction of real-time signals further complicates the paradox: a negative OFI (buying pressure) justifies acceleration; a positive OFI (selling pressure) justifies deceleration. These dynamic adjustments, impossible for TWAP and VWAP, are precisely what RL agents learn through repeated market interaction.

The signal-conditional execution problem was studied by Bertsimas & Lo (1998) and later by Cartea, Donnelly & Jaimungal (2018), who show that an agent aware of short-term price direction can reduce IS by 20-40% relative to signal-blind strategies. Their semi-analytical solutions assume known signal dynamics, however, an assumption that RL relaxes by learning the signal-action relationship from data. This 'model-free adaptation' is a key advantage of RL over classical stochastic control in the context of real markets where signal dynamics are non-stationary and partially unknown. Our empirical results, where PPO learns to exploit OFI signals without an explicit signal model, illustrate this advantage concretely.

### *1.3.5 Price Discovery in Fragmented European Markets*

Market fragmentation fundamentally alters the price discovery process, the mechanism by which dispersed private information is aggregated into publicly observable prices. Hasbrouck (1995) developed the information share concept: the fraction of total price variance innovation attributable to each venue's quotes. Applied to European markets by De Jong & Schotman (2010), this methodology consistently finds that primary exchanges (Euronext Paris for French stocks) contribute 55-75% of price discovery despite holding 30-40% of volume, reflecting the concentration of informed institutional flow on regulated markets.

The practical implication for execution is that large institutional orders on the primary exchange send stronger informational signals than equivalent orders on MTFs. A sophisticated agent, as a well-trained RL agent should be, will route information-revealing slices (large quantities) to venues where adverse selection risk is lower, while concentrating informationally-neutral volume on the primary exchange. Menkveld (2013) documented HFT market makers as the synchronisation mechanism ensuring cross-venue price consistency within milliseconds. Their presence means that impact on one venue propagates instantly to all venues through quote adjustment, a critical bound on the fragmentation alpha we measure empirically (35% of total gains), consistent with the partial but not complete exploitability of cross-venue inefficiencies.

## 1.6 Synthesis and Research Gap

The four literature axes reviewed in this chapter converge on a clear gap: no published study has applied Deep Reinforcement Learning to optimal execution on real, fragmented, multi-venue European equity markets with venue-specific impact calibration and a cross-venue allocation action space. Our research fills this gap through a full empirical pipeline from real Level 2 data to trained agents to rigorous out-of-sample evaluation. The conceptual framework of Chapter 2 translates the four literature strands, IS decomposition, optimal execution models, European market microstructure, Deep RL, into a unified and testable MDP formulation.

## 3.5 Training Infrastructure and Implementation Details

Training the DQN and PPO agents requires substantial computational infrastructure. The pipeline was implemented in Python (PyTorch for neural networks) with a custom execution environment built on top of the Level 2 data. Key implementation choices: vectorised parallel episode simulation for efficient data collection; automatic mixed precision training for GPU acceleration; gradient clipping (max norm 0.5) to prevent training instabilities; and checkpoint-based training with periodic validation on the held-out set for early stopping.

Each agent was trained for 500,000 environment steps on the 2022-2023 training set, requiring approximately 24 hours on an NVIDIA A100 GPU. Three independent runs per algorithm with different random seeds confirmed convergence robustness: cross-seed standard deviation of validation IS was ±0.4 bp for PPO and ±0.6 bp for DQN, indicating stable convergence. The reported results use the seed achieving the best validation IS.

## 3.6 Data Preprocessing Pipeline

Raw tick-by-tick Level 2 data required a five-stage preprocessing pipeline. First, data cleaning: removal of erroneous quotes (bid > ask, zero-quantity levels, disconnected timestamps) and venue identifier standardisation. Second, time alignment: projection of all venue snapshots onto a common one-minute CET clock using last-value interpolation. Third, ADV estimation: rolling 20-day average daily volume per stock per venue, updated daily. Fourth, feature computation: OFI, per-venue depth, and inter-venue spreads computed from aligned snapshots. Fifth, staleness filtering: quotes unchanged for more than 30 minutes flagged as stale and replaced with zero in the depth feature, affecting 3-5% of observations, principally on lower-activity venues.

Quote staleness is an important data quality issue specific to multi-venue data that is not discussed in the single-venue literature. In our dataset, Aquis Exchange shows the highest incidence of stale quotes (up to 8% of observations during low-activity periods), reflecting its smaller market share and sparser quote update frequency relative to CBOE Europe and Euronext Paris. Our conservative zero-replacement policy for stale quotes means the agent slightly underestimates available liquidity on lower-activity venues, a bias that may modestly understate the true fragmentation alpha. Future work should explore more sophisticated staleness detection using order book event sequencing rather than purely time-based filters.

## 4.4 Cross-Sectional Analysis by Stock Characteristics

Cross-sectional regression analysis relates per-stock percentage IS reductions (PPO vs. TWAP) to four stock-level characteristics: log market capitalisation, log ADV, HHI of liquidity fragmentation, and average daily volatility. The regression reveals that HHI is the strongest predictor of RL performance gain (coefficient −0.42, $t = -5.1$, $p < 0.001$): more fragmented stocks yield larger percentage IS reductions, directly supporting H2's mechanism. Log ADV is the second most significant predictor (coefficient 0.18, $t = 2.8$, $p = 0.008$): higher-volume stocks show larger gains, consistent with the richer order flow signal for training. Market capitalisation and volatility show weaker associations after controlling for HHI and ADV.

These cross-sectional results have a direct practical implication: RL-based execution should be prioritised for large-cap, highly-liquid, highly-fragmented stocks, the core holdings of major European institutional portfolios. A tiered deployment approach, RL for the top quartile of stocks by HHI-adjusted liquidity, Almgren-Chriss for the remainder, would capture the majority of the total economic benefit while limiting implementation complexity and model maintenance overhead.

## 4.5 Intraday Analysis and Time-of-Day Effects

Analysis of IS reductions by execution start time (9:30–15:30) reveals that PPO's advantage over TWAP is largest for mid-session executions (11:00-13:00), where the low-liquidity trough creates the greatest distortion from TWAP's uniform schedule. The average IS advantage in this window is approximately 12 bp (PPO 10.2 bp vs. TWAP 22.1 bp), compared with 8 bp for morning starts and 9 bp for afternoon starts. This finding is practically important: large executions during mid-session, common in transition management and index replication contexts, benefit most from RL adoption.

The intraday pattern of multi-venue allocation reveals further adaptive behaviour. During the early continuous session (9:30-10:00), the agent concentrates execution on Euronext Paris, consistent with lower competing activity on MTFs at market open. During mid-session continuous trading, cross-venue allocation shifts dynamically with OFI and depth signals as shown in Figure 9. This intraday adaptation pattern is not programmable in fixed-rule SOR systems and represents a qualitatively distinct capability of the RL approach.

## 4.6 Cost-Benefit Analysis of RL Deployment

The economic case for deploying RL-based execution requires comparing expected IS savings against full development and operational costs. For a fund managing €5 billion in European equities with 40% annual turnover (€2 billion traded volume), the expected annual IS saving from RL versus Almgren-Chriss is 0.9 bp × €2 billion × 0.5 (regime-conditional activation fraction) = approximately €900,000 per year. Against TWAP (a more typical current baseline), the saving is 9.4 bp × €2 billion × 0.5 ≈ €9.4 million per year.

Development and infrastructure costs: model development (€50,000-150,000 one-time), Level 2 data (€30,000-80,000/year), connectivity infrastructure (€100,000-300,000 one-time + €50,000-100,000/year), ongoing maintenance (€15,000-30,000/year). Total first-year cost: €200,000-650,000; ongoing annual cost: €100,000-310,000. ROI ranges from 40% (conservative, vs. Almgren-Chriss) to 4,700% (vs. TWAP), with payback periods of less than one year under any reasonable assumption. The economic case is compelling for funds managing €2 billion or more in European equities. For smaller funds, broker-offered RL execution services may be more cost-effective than proprietary development.

# Appendices

## Appendix A, Interview Guide (Qualitative Triangulation)

Six semi-structured interviews were conducted with execution practitioners between September and November 2026: two heads of algorithmic execution at European investment banks, two quantitative portfolio managers at institutional asset managers, and two senior engineers at algorithmic brokers. Interviews lasted 45-75 minutes. The following guide was used:

- Can you describe the algorithmic execution decision process within your organisation? What criteria drive algorithm selection for a given order?
- Which execution algorithms dominate your practice (TWAP, VWAP, IS, adaptive)? Under what circumstances do you switch between them?
- How do you manage liquidity fragmentation across European venues? Proprietary SOR or broker SOR?
- Have you experimented with ML or RL approaches for execution? What outcomes have you observed?
- What are the main technical, operational and regulatory barriers to RL deployment in production?
- How do you satisfy MiFID II best execution reporting requirements with adaptive or ML algorithms?
- Do you use Implementation Shortfall as your primary execution quality metric? What complementary indicators do you monitor?

## Appendix B, RL Agent Hyperparameters

### Table B.1, DQN Hyperparameters

- Architecture: 3 fully-connected layers (128-64-32 neurons), ReLU activation, Xavier initialisation
- Replay buffer: 100,000 transitions, uniform sampling
- Mini-batch size: 64 transitions per gradient step
- Discount factor $\gamma$: 0.99
- Learning rate: $3\times10^{-4}$ (Adam, $\beta_1$=0.9, $\beta_2$=0.999, $\varepsilon$=$10^{-8}$)

- Target network update: every 500 gradient steps (hard update)
- Exploration: ε-greedy, ε linearly decayed from 1.0 to 0.01 over 50,000 steps
- Total training steps: 500,000 (approximately 250 complete episode sweeps)

**Table B.2, PPO Hyperparameters**

- Architecture: shared 3-layer backbone (128-64-32), separate linear actor and critic heads
- Clip ratio ε: 0.2
- Entropy coefficient: 0.01
- Value loss coefficient: 0.5
- Gradient clipping: max norm 0.5
- Optimisation epochs per rollout: 10
- Mini-batch size: 256 (from 2,048-step rollout buffer)
- Rollout horizon: 2,048 environment steps
- Learning rate: $3\times10^{-4}$ (Adam), linearly annealed to 0 over training
- Advantage normalisation: batch mean-zero, unit-variance

## Appendix C, List of Figures



## Appendix D, List of Tables

## Appendix J, Research Context and Institutional Acknowledgements

This dissertation was conducted within the dual institutional context of the ESLSCA MBA programme in Trading and Financial Markets and the author's professional practice at Tikehau Capital, an alternative asset management firm headquartered in Paris with EUR 45 billion in assets under management. The research question emerged directly from the author's daily professional confrontation with the challenges of institutional execution on fragmented European equity markets. This dual grounding, academic rigour and professional relevance, shaped both the research design and the managerial orientation of the findings.

The research was conducted independently from Tikehau Capital's proprietary trading activities. All data was obtained through commercial licensing agreements independent of the firm, and no proprietary information was incorporated in any aspect of the research. The dissertation supervisor, Abdelbaric EL ALAOUI, provided guidance ensuring the research design meets the standards of the quantitative finance academic community. External perspectives from engagement with practitioners at the Risk.net Quant Congress Europe (2024) and the ESMA Academic Research Workshop (2024) helped calibrate the practical relevance of the findings.

Looking ahead, several natural extensions of this research, including the multi-asset execution framework, the online learning agent for non-stationary adaptation, and the adversarial robustness analysis of RL patterns versus HFT market maker responses, are planned as post-graduation projects, building on the empirical infrastructure developed for this dissertation. The rapidly evolving regulatory landscape for algorithmic trading in Europe, including the ongoing MiFID review and the Capital Markets Union initiative, will continue to shape the practical context for RL execution research in ways that make sustained engagement with both the academic and regulatory communities essential for researchers and practitioners seeking to advance this field.

The implementation of RL-based execution algorithms at institutional scale represents one of the most concrete and measurable applications of artificial intelligence to professional finance practice. Unlike many AI applications in finance where the causal mechanisms are opaque and the empirical validation is difficult, the execution domain offers a clear objective function (IS minimisation), rich real-world data for validation, and direct economic stakes that provide strong incentive for rigorous evaluation. This combination of theoretical clarity, empirical tractability, and practical relevance makes optimal execution an ideal laboratory for studying the conditions under which AI delivers genuine and durable value in financial markets, a question whose answer will shape the trajectory of institutional investment management for decades to come.

The author expresses sincere gratitude to Abdelbaric EL ALAOUI for the intellectual guidance and constructive feedback provided throughout the supervision of this dissertation, to the six practitioners who generously agreed to participate in the qualitative interviews and shared their operational expertise, and to the academic and professional colleagues whose observations and critiques substantially strengthened the research. Any remaining errors of fact, interpretation, or analysis are the sole responsibility of the author. This dissertation is submitted in partial fulfilment of the requirements for the Master of Business Administration degree in Trading and Financial Markets at ESLSCA Business School Paris, academic year 2025-2026. Word count excluding tables, figures, references, and appendices: approximately 28,000 words.

The quantitative finance industry stands at an inflection point with respect to the integration of machine learning methods into core trading and investment management processes. Implementation Shortfall optimisation represents one of the most tractable and directly measurable domains for this integration, and the results of this dissertation suggest that the technology readiness for RL-based execution is higher than commonly assumed. The primary barriers to adoption are no longer technical, the algorithms work and the data is available, but organisational and regulatory: the development of compliance frameworks, the cultivation of internal expertise, and the alignment of incentives between execution desks, risk management, and senior management are the bottlenecks that industry leaders must address in the near term.

The broader implications of this research extend beyond the specific IS reduction numbers to a more fundamental question about the future of institutional investment management. If execution algorithms become increasingly sophisticated and differentiated through machine learning, execution quality will increasingly serve as a competitive differentiator between asset managers competing for the same institutional mandates. Managers who invest early in RL execution capabilities, building the data infrastructure, the talent, and the regulatory frameworks required, will have a structural cost advantage over laggards. Conversely, managers who outsource execution entirely to broker algorithms will find themselves at the mercy of the broker's incentives and capabilities, with limited visibility into whether best execution is truly being achieved.

The research findings also carry implications for European financial market structure and policy. The identification of fragmentation as a source of exploitable execution alpha suggests that the current multi-venue European equity market structure creates differentials in execution quality between sophisticated institutions capable of exploiting cross-venue dynamics and less sophisticated ones that cannot. This differential is not inherently problematic from a market efficiency perspective, competitive advantages based on technology investment and expertise are a normal feature of financial markets. However, it does raise questions about the adequacy of current MiFID II best execution requirements in ensuring a level playing field for retail investors and smaller institutional investors who lack the resources to develop RL execution capabilities.

The development of a European Consolidated Tape, a single, authoritative source of real-time and historical European equity trading data, would be an important step toward democratising access to the multi-venue data infrastructure that underlies RL execution optimisation. By reducing the information asymmetry between large institutions with direct multi-venue data feeds and smaller institutions relying on delayed or aggregated data, the Consolidated Tape would lower the barriers to RL adoption for a broader range of market participants. Our research suggests that the economic benefits of such a policy measure, measured in terms of aggregate IS reduction across European institutional investors, could be substantial.

In conclusion, this dissertation has demonstrated that Reinforcement Learning offers a statistically robust, economically significant, and practically implementable improvement in execution quality on fragmented European equity markets. The three research hypotheses, RL superiority over classical benchmarks, fragmentation as exploitable alpha, and regime conditionality of performance, have been validated with high statistical confidence on real European market data. The managerial recommendations, regulatory framework, and implementation roadmap provide a concrete path for institutional practitioners to translate these research findings into operational reality. The broader research agenda opened by this work, multi-asset extensions, online learning, adversarial robustness, regulatory adaptation, ensures that optimal execution in fragmented European markets will remain a productive and practically relevant domain for academic and applied research for years to come.

The author declares no conflicts of interest in connection with this research. The views expressed in this dissertation are those of the author alone and do not represent the views of Tikehau Capital, ESLSCA Business School Paris, or any other institution with which the author is or has been affiliated. This work is dedicated to all those who believe that rigorous, practically grounded research can improve the functioning of financial markets for the benefit of all participants.